Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit.**

Vintage mix â€" read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

The one rule behind this layout

The rung is a property of (CARD, COLUMN) â€" never of the column name. `beta` is edge-rung on the edge card and family-rung on the gene_family card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits â€" otherwise `n` and `z` match ordinary prose.

One correction the measurement needed: mention-frequency is not importance â€" a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

Cards

- `edge â€" 186 columns, 45 blocks`
- `gene â€" 135 columns, 29 blocks`
- `gene_family â€" 61 columns, 12 blocks`
- `arm â€" 296 columns, 43 blocks`
- `seed_family â€" 33 columns, 2 blocks`

realization/edge_card.tsv

edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer â€” check the vintage banner before quoting a fitted one.

186 columns **5,649** rows **45** blocks median coverage **72%**

Contents â€” 45 blocks, most important first

(BARE) Â· 11

core â€” the columns findings rest on

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.

11 columns Â· coverage **63%â€“100%** (median 96%) Â· **3** share the block description
4× real, signed Â· 2× identifier Â· 2× real â‰¥ 0 Â· 2× signed âˆ’1â€¦1 Â· 1× boolean

arm
KEY IDENTIFIER 100%

KEY â€” mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED

beta
EDGE REAL Â‰¥ 0 100%

EDGE-rung coupling coefficient â€” `readouts.run(level='arm')`, the same-seed collapse REMOVED. AUTHORED

dGlobal_HLY_NAT
`ARM` `REAL, SIGNED` `63%`

in the GTEx-healthy state Â· in matched normal (NAT) `DERIVED`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

dGlobal_HLY_TUM
`ARM` `REAL, SIGNED` `63%`

in the GTEx-healthy state Â· in tumour `DERIVED`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

dGlobal_NAT_TUM
`ARM` `REAL, SIGNED` `68%`

in matched normal (NAT) Â· in tumour `DERIVED`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

dShare_M_own
`EDGE` `SIGNED Â^'1Â€¦1` `65%`

Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT. `AUTHORED`

---

dShare_raw_own
`EDGE` `SIGNED Â^'1Â€¦1` `65%`

As `dShare_M_own` but on RAW abundance share, with the model's `M` weights removed. `AUTHORED`

---

gene
`KEY` `IDENTIFIER` `100%`

KEY â€" HGNC gene symbol. `AUTHORED`

---

identified
`EDGE` `BOOLEAN` `100%`

`|z| > 2` on the UNCALIBRATED posterior SD, where `z = beta / beta_sd` (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). **True on 24.8% of edges.** `AUTHORED`

---

identity
`EDGE` `REAL, SIGNED` `96%`

Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights. `AUTHORED`

---

z
`EDGE` `REAL Â%¥ 0` `100%`

`beta / beta_sd` for the EDGE-level fit (`readouts.run(level='arm')` , the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD. `AUTHORED`

COUPLING_ Â· 13

core â€" the columns findings rest on

> Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
>
> **13** columns Â· coverage **3%â€"100%** (median 62%) Â· **9** share the block description
> 5× signed â^'1â€¦1 Â· 5× p-value Â· 3× real, signed

coupling_fam
`FAMILY` `SIGNED Â^'1Â€¦1` `100%`

Family-level coupling for the edge's (gene, seed_family) cell. `AUTHORED`

---

coupling_hly
`EDGE` `SIGNED Â^'1Â€¦1` `56%`

in the GTEx-healthy state `DERIVED`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

coupling_hly_surrogate
`EDGE` `SIGNED Â^'1Â€¦1` `3%`

Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal. `AUTHORED`

---

coupling_nat
`EDGE` `SIGNED Â^'1Â€¦1` `62%`

in matched normal (NAT) `DERIVED`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

coupling_p_hly
`EDGE` `P-VALUE` `56%`

p-value Â· in the GTEx-healthy state `DERIVED`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

coupling_p_hly_resolved
EDGE P-VALUE 59%

p-value Â· in the GTEx-healthy state DERIVED

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€" 3.6% of edges*

---

coupling_p_hly_surrogate
EDGE P-VALUE 3%

p-value Â· in the GTEx-healthy state DERIVED

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€" 3.6% of edges*

---

coupling_p_nat
EDGE P-VALUE 62%

p-value Â· in matched normal (NAT) DERIVED

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

coupling_p_tum
EDGE P-VALUE 69%

p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; <0.05 on 16.2%. AUTHORED

---

coupling_tum
EDGE SIGNED Âˆ'1Â€¦1 72%

in tumour DERIVED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

coupling_z_hly
EDGE REAL, SIGNED 59%

z-score Â· in the GTEx-healthy state DERIVED

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

coupling_z_nat
EDGE REAL, SIGNED 62%

z-score Â· in matched normal (NAT) DERIVED

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

coupling_z_tum
EDGE REAL, SIGNED 69%

The tumour coupling expressed in NULL SDs â€" the CALIBRATED effect size, and the scale to quote. Median **âˆ'0.32**; |z|>2 on 15.5%, |z|>3 on 6.4%. AUTHORED

SEED_ Â· 1

core â€" the columns findings rest on

1 columns Â· coverage **100%â€“100%** (median 100%)
1× identifier

seed_family
FAMILY IDENTIFIER 100%

KEY â€" the seed family itself, one row per family. AUTHORED

ARM_ Â· 15

core â€" the columns findings rest on

15 columns Â· coverage **20%â€“100%** (median 69%)
4× real â‰¥ 0 Â· 4× real, signed Â· 3× fraction 0â€“1 Â· 2× categorical Â· 1× percent Â· 1× boolean

arm_credit_share
ARM-IN-FAMILY FRACTION 0Â€"1 20%

This arm's share of its (gene, seed_family) cell's credit. âœ… Verified to sum to 1.0000 within every one of the 467 multi-arm cells, and it genuinely varies PER ARM inside the cell (467/467). AUTHORED

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€" 20.3% of edges*

---

arm_dbeta
FAMILY REAL Â‰¥ 0 20%

maxâˆ'min Î² across the cell's member arms. AUTHORED

---

arm_detection
ARM FRACTION 0Â€"1 95%

Fraction of patients in which this ARM is measured at all (`readouts._detection`). AUTHORED

---

arm_id_status
ARM-IN-FAMILY CATEGORICAL 100%

How the arm's identity within its cell was resolved: `resolved_by_dose` 2,344 Â· `singleton` 2,260 Â· `unvalidated_balanced` 1,045. AUTHORED

---

arm_iqr
`ARM` `REAL Â‰¥ 0` `69%`

Interquartile range of the arm's abundance â€" its DYNAMIC RANGE. AUTHORED

---

arm_lfc_HLY_NAT_raw
`ARM` `REAL, SIGNED` `63%`

log-FC GTEx-healthy â†' TCGA-NAT, raw. AUTHORED

---

arm_lfc_HLY_TUM_QN
`ARM` `REAL, SIGNED` `64%`

log-FC GTEx-healthy â†' TCGA-tumour, on quantile-normalised values. AUTHORED

---

arm_lfc_HLY_TUM_raw
`ARM` `REAL, SIGNED` `63%`

As `arm_lfc_HLY_TUM_QN` but against the RAW GTEx median. AUTHORED

---

arm_lfc_NAT_TUM
`ARM` `REAL, SIGNED` `68%`

log-FC of the arm's median abundance, matched NAT â†' tumour. AUTHORED

---

arm_med_rpm
`ARM` `REAL Â‰¥ 0` `69%`

Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 â€" the gap is arms with no expression record, not zeros. AUTHORED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_pct_above_floor
`ARM` `PERCENT` `69%`

Percent of samples in which the arm sits above the detection floor (median 99). Feeds `arm_spiker` together with `arm_iqr`. AUTHORED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_role_in_family
`ARM-IN-FAMILY` `CATEGORICAL` `72%`

â›" **The ARM's DOSE role inside its seed family â€" NOT the family's role in the gene**, which is what the name suggests. AUTHORED

---

arm_sep_z
`FAMILY` `REAL Â‰¥ 0` `20%`

`arm_dbeta` in units of the cell's own pooled posterior SD â€" can these same-seed arms be told apart at all. AUTHORED

---

arm_share_of_family_dose
`ARM-IN-FAMILY` `FRACTION 0â€"1` `69%`

The arm's share of its seed family's TOTAL dose â€" denominator is the **full family**, not the design (median 0.69). The quantity `family_role` is cut from. AUTHORED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_spiker
`ARM` `BOOLEAN` `69%`

`arm_pct_floor < 40 AND arm_iqr > 1.5` â€" the arm sits at/below the detection floor in most samples yet swings widely when present. AUTHORED

ADM_ Â· 4

core â€" the columns findings rest on

> **4** columns Â· coverage **87%â€"87%** (median 87%)
> 4× boolean

adm_admissible
`EDGE` `BOOLEAN` `87%`

The admissibility verdict: site + evidence + expression. AUTHORED

---

adm_expr_gtex
`EDGE` `BOOLEAN` `87%`

ADMISSIBILITY â€" could this edge repress in breast AT ALL? `has_site` (seed match in the 3'UTR) AND `expressed` (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED

---

adm_expr_tcga
`EDGE` `BOOLEAN` `87%`

Is the arm expressed above floor in TCGA? AUTHORED

---

adm_has_site
`EDGE` `BOOLEAN` `87%`

Does the arm have a seed match in this target's 3'UTR. AUTHORED

BETA_ Â· 8

core â€” the columns findings rest on

8 columns Â· coverage **20%â€“100%** (median 100%)
4× real â‰¥ 0 Â· 4× fraction 0â€“1

beta_arm
EDGE | REAL Â‰¥ 0 | 20%

The arm's OWN Î² from the within-cell fit (`arm_rung.py`), defined only in multi-arm cells (fill 0.203). AUTHORED

beta_deconv
EDGE | REAL Â‰¥ 0 | 100%

Î² refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out). AUTHORED

beta_frac
EDGE | FRACTION 0Â€“1 | 100%

MAGNITUDE share (beta_f / sum beta). AUTHORED

beta_frac_abs
EDGE | FRACTION 0Â€“1 | 100%

|E[Î²_f]| / Î£|E[Î²]| â€” the share computed from the POINT ESTIMATES. AUTHORED

beta_frac_deconv
EDGE | FRACTION 0Â€“1 | 100%

`beta_frac` recomputed on the deconv design â€” the composition-adjusted share of the gene's budget. AUTHORED

beta_frac_sd
EDGE | FRACTION 0Â€“1 | 100%

Posterior SD of the per-draw fraction Î²_f/Î£Î². AUTHORED

beta_sd
EDGE | REAL Â‰¥ 0 | 100%

Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta. AUTHORED

beta_sd_deconv
EDGE | REAL Â‰¥ 0 | 100%

Posterior SD on the deconv design. AUTHORED

N_ Â· 5

core â€” the columns findings rest on

5 columns Â· coverage **20%â€“100%** (median 100%)
4× count Â· 1× constant

n_HLY_meas
EDGE | COUNT | 72%

How many of this gene's arms have a MEASURED healthy (GTEx) level â€” as opposed to one manufactured by the multi-mapping collapse. AUTHORED

n_arm_in_cell
FAMILY | COUNT | 20%

How many arms sit in this edge's (gene, seed_family) CELL â€” i.e. how many of that family's members appear in THIS gene's design. AUTHORED

n_fam
GENE | COUNT | 100%

How many families compete at this edge's GENE. Gene-rung â€” it repeats across the gene's edges by construction. beta is mathematically inert where it is 1. AUTHORED

n_family_in_design
FAMILY | COUNT | 100%

Members of this arm's seed family **THAT ARE IN THIS GENE'S DESIGN** â€” 1 on **4,497 of 5,644 edges (79.7%)**. AUTHORED

n_samples
GENE | CONSTANT | 100%

Samples the edge was fit on â€” CONSTANT (1,040) across the card. AUTHORED

SHARE_ Â· 4

core â€” the columns findings rest on

The arm's share of the gene's total regulatory dose in one state (HLY / NAT / TUM). Within-gene, sums to 1 across the gene's arms.

4 columns Â· coverage **46%â€“72%** (median 72%) Â· **2** share the block description

4× fraction 0â€“1

share_HLY
EDGE | FRACTION 0Â€“1 | 72%

The arm's share of the gene's healthy (GTEx) pressure budget, median **0.129**. AUTHORED

share_HLY_meas
EDGE | FRACTION 0Â€“1 | 46%

`share_HLY` over ONLY the arms GTEx genuinely measures â€” invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median **0.244** against the imputed 0.129. AUTHORED

share_NAT
EDGE | FRACTION 0Â€“1 | 72%

in matched normal (NAT) DERIVED

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

share_TUM
EDGE | FRACTION 0Â€“1 | 72%

in tumour DERIVED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

RANK_ Â· 4

core â€” the columns findings rest on

The arm's WITHIN-GENE rank by dose in one state.

4 columns Â· coverage **46%â€“72%** (median 72%) Â· **4** share the block description

4× count

rank_HLY
EDGE | COUNT | 72%

in the GTEx-healthy state DERIVED

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

rank_HLY_meas
EDGE | COUNT | 46%

in the GTEx-healthy state Â· measured only (un-imputed) DERIVED

> *Defined on: edges whose gene has â‰¥1 GTEx-MEASURED regulator â€” âš NaN = GTEx cannot see it (abstention), NOT 0*

rank_NAT
EDGE | COUNT | 72%

in matched normal (NAT) DERIVED

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

rank_TUM
EDGE | COUNT | 72%

in tumour DERIVED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

SHIFT_ Â· 1

core â€” the columns findings rest on

1 columns Â· coverage **72%â€“72%** (median 72%)

1× categorical

shift_class
EDGE | CATEGORICAL | 72%

Cross-state class for the edge. AUTHORED

CAL_ Â· 7

core â€” the columns findings rest on

CALIBRATED posterior width (`learned/calibration.py`): the reported SD corrected for the measured 1.18â€“1.34x narrowness, with `_lo` / `_hi` spanning the correction constant's own +/-1 SD.

7 columns Â· coverage **100%â€“100%** (median 100%) Â· **6** share the block description
4× real â‰¥ 0 Â· 3× boolean

cal_beta_sd
EDGE | REAL Â‰¥ 0 | 100%

Î² Â· standard deviation DERIVED

cal_beta_sd_hi
EDGE | REAL Â‰¥ 0 | 100%

Î² Â· standard deviation DERIVED

cal_beta_sd_lo
EDGE | REAL Â‰¥ 0 | 100%

Î² Â· standard deviation DERIVED

cal_identified
EDGE | BOOLEAN | 100%

Is the edge identified (`|cal_z| > 2`) under the CALIBRATED posterior width â€” the honest version of `identified`: **1,110 vs 1,375 edges**, i.e. **19.9% [18.1â€“22.4]** against 24.8%. AUTHORED

cal_identified_hi
EDGE | BOOLEAN | 100%

CALIBRATED posterior width (`learned/calibration.py`): the reported SD corrected for the measured 1.18â€“1.34x narrowness, with `_lo` / `_hi` spanning the correction constant's own +/-1 SD. BLOCK-LEVEL

cal_identified_lo
EDGE | BOOLEAN | 100%

CALIBRATED posterior width (`learned/calibration.py`): the reported SD corrected for the measured 1.18â€“1.34x narrowness, with `_lo` / `_hi` spanning the correction constant's own +/-1 SD. BLOCK-LEVEL

cal_z
EDGE | REAL Â‰¥ 0 | 100%

z-score DERIVED

ECHIM_ Â· 3

core â€” the columns findings rest on

CHIMERIC ligation evidence at its NATIVE (gene, arm) rung â€” a physical duplex read, per source, never pooled.

3 columns Â· coverage **8%â€“26%** (median 22%) Â· **2** share the block description
2× count Â· 1× real â‰¥ 0

echim_manakov_w
EDGE | REAL Â‰¥ 0 | 22%

CHIMERIC ligation evidence at its NATIVE (gene, arm) rung â€” a physical duplex read, per source, never pooled. BLOCK-LEVEL

> *Defined on: edges with a chimeric (AGO-ligation) duplex â€” âš per-source, NEVER pooled; cross-tissue, not breast*

echim_n_sources
EDGE | COUNT | 26%

How many chimeric-eCLIP sources support this edge (1 on 1,151 Â· 2 on 262 Â· 3 on 34). AUTHORED

echim_tarbase_w
EDGE | COUNT | 8%

CHIMERIC ligation evidence at its NATIVE (gene, arm) rung â€” a physical duplex read, per source, never pooled. BLOCK-LEVEL

> *Defined on: edges with a chimeric (AGO-ligation) duplex â€” âš per-source, NEVER pooled; cross-tissue, not breast*

CTX_ Â· 18

supporting â€” read these when the core raises a question

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

18 columns Â· coverage **84%â€“100%** (median 99%) Â· **12** share the block description
4× real â‰¥ 0 Â· 4× signed âˆ’1â€¦1 Â· 3× real, signed Â· 3× count Â· 2× boolean Â· 1× categorical Â· 1× fraction 0â€“1

ctx_apriori_class
GENE · CATEGORICAL · 100%
class label DERIVED

ctx_arm_abundant
ARM · BOOLEAN · 95%
Is this ARM above the design's abundance threshold. AUTHORED

ctx_arm_dose
ARM · REAL Â‰¥ 0 · 95%
The ARM's dose in this gene's design context. AUTHORED

ctx_ceiling
GENE · SIGNED Âˆ’1Â€¦1 · 99%
Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. BLOCK-LEVEL

ctx_collapse
GENE · REAL Â‰¥ 0 · 100%
Mean arms collapsed per family in this gene's design (median 1.07) â€” how much the seed-family collapse actually merged. Near 1 means the collapse was almost a no-op for this gene. AUTHORED

ctx_d_collin
GENE · SIGNED Âˆ’1Â€¦1 · 84%
Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap. AUTHORED

> *Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)*

ctx_d_fam
GENE · REAL, SIGNED · 99%
Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families). AUTHORED

ctx_dose_max
GENE · REAL Â‰¥ 0 · 100%
dose Â· maximum DERIVED

ctx_dose_med
GENE · REAL Â‰¥ 0 · 100%
dose Â· median DERIVED

ctx_frac_abund
GENE · FRACTION 0Â€“1 · 100%
fraction Â· unweighted abundance-sum reference DERIVED

ctx_gap_core
GENE · SIGNED Âˆ’1Â€¦1 · 99%
gap Â· core confounder block DERIVED

ctx_gap_d_dose
GENE · REAL, SIGNED · 99%
Dose asymmetry between the real and decoy designs (median âˆ’0.43). AUTHORED

ctx_gap_deconv
GENE · SIGNED Âˆ’1Â€¦1 · 99%
gap Â· composition-adjusted (deconvolution block) DERIVED

ctx_gap_retention
GENE · REAL, SIGNED · 95%
gap Â· retention DERIVED

ctx_measurable
GENE · BOOLEAN · 99%
Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. BLOCK-LEVEL

ctx_n_abund
GENE · COUNT · 100%

count Â· unweighted abundance-sum reference DERIVED

ctx_n_fam_abund
GENE · COUNT · 100%

count Â· per seed family Â· unweighted abundance-sum reference DERIVED

ctx_n_fam_multi
GENE · COUNT · 100%

count Â· per seed family DERIVED

HEALTHY_ Â· 3

supporting â€" read these when the core raises a question

> Healthy-state legs and flags â€" including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim.
>
> **3** columns Â· coverage **64%â€"72%** (median 64%) Â· **3** share the block description
>
> 1× categorical Â· 1× real â‰¥ 0 Â· 1× boolean

healthy_leg
ARM · CATEGORICAL · 72%

Healthy-state legs and flags â€" including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim. BLOCK-LEVEL

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

healthy_potential
ARM · REAL â‰¥ 0 · 64%

Healthy-state legs and flags â€" including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim. BLOCK-LEVEL

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

healthy_uninformative
ARM · BOOLEAN · 64%

Healthy-state legs and flags â€" including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim. BLOCK-LEVEL

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

IDENTITY_ Â· 5

supporting â€" read these when the core raises a question

> **5** columns Â· coverage **96%â€"98%** (median 96%)
>
> 2× fraction 0â€"1 Â· 2× real, signed Â· 1× boolean

identity_abs
EDGE · FRACTION 0â€"1 · 96%

|identity| as a fraction of the gene's total |identity|, **clipped to [0,1]**. AUTHORED

identity_allocated
EDGE · REAL, SIGNED · 96%

Family identity with phantom minor arms forced to 0 â€" the allocation actually used downstream. AUTHORED

identity_coherence
EDGE · FRACTION 0â€"1 · 96%

`1 / Î£|identity|` over the gene, clipped to (0,1] â€" how much the gene's identity shares CANCEL. AUTHORED

identity_deconv
EDGE · REAL, SIGNED · 98%

Shapley identity recomputed under the composition-adjusted design. AUTHORED

identity_reliable
EDGE · BOOLEAN · 96%

Is this edge's Shapley `identity` trustworthy â€" `identity_coherence` â‰¥ 0.5 AND |identity| â‰¤ 1? AUTHORED

DOSE_ Â· 7

supporting â€" read these when the core raises a question

> Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.
>
> **7** columns Â· coverage **55%â€"72%** (median 72%) Â· **6** share the block description
>
> 3× count Â· 2× real â‰¥ 0 Â· 1× categorical Â· 1× constant

dose_class EDGE CATEGORICAL 72%
class label DERIVED
Defined on: edges present in the progression panel (arm expressed + state measured)

dose_comp_retention ARM REAL Â‰¥ 0 55%
retention DERIVED
Defined on: edges with a matched NAT leg (n~104 paired participants)

dose_confounded ARM CONSTANT 72%
Is this edge's NATâ†'tumour dose a composition/proliferation artifact (either gated retention < 0.4)? AUTHORED

dose_prolif_retention ARM REAL Â‰¥ 0 55%
retention DERIVED
Defined on: edges with a matched NAT leg (n~104 paired participants)

dose_rank_HLY EDGE COUNT 72%
rank Â· in the GTEx-healthy state DERIVED
Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)

dose_rank_NAT EDGE COUNT 72%
rank Â· in matched normal (NAT) DERIVED
Defined on: edges with a matched NAT leg (n~104 paired participants)

dose_rank_TUM EDGE COUNT 72%
rank Â· in tumour DERIVED
Defined on: edges present in the progression panel (arm expressed + state measured)

D_ Â· 4

supporting â€" read these when the core raises a question

> Change in the arm's WITHIN-GENE dose rank between two states.
>
> **4** columns Â· coverage **46%â€"72%** (median 72%) Â· **3** share the block description
>
> 4× real, signed

d_rank_HLY_NAT EDGE REAL, SIGNED 72%
rank Â· in the GTEx-healthy state Â· in matched normal (NAT) DERIVED
Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)

d_rank_HLY_TUM EDGE REAL, SIGNED 72%
rank Â· in the GTEx-healthy state Â· in tumour DERIVED
Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)

d_rank_HLY_TUM_meas EDGE REAL, SIGNED 46%
`d_rank_HLY_TUM` recomputed over arms with a MEASURED healthy level only â€" the collapse-safe version. Prefer it over the bare column. AUTHORED
Defined on: edges whose gene has â‰¥1 GTEx-MEASURED regulator â€" âš NaN = GTEx cannot see it (abstention), NOT 0

d_rank_NAT_TUM EDGE REAL, SIGNED 72%
rank Â· in matched normal (NAT) Â· in tumour DERIVED
Defined on: edges with a matched NAT leg (n~104 paired participants)

GENE_ Â· 6

supporting â€" read these when the core raises a question

> Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.
>
> **6** columns Â· coverage **72%â€"100%** (median 99%) Â· **5** share the block description
> 2× categorical Â· 2× boolean Â· 1× real â‰¥ 0 Â· 1× real, signed

gene_cancer_role
GENE · CATEGORICAL · 100%

â›" **NOT the arm's role â€" the GENE's cancer role** (`oncogene` / `tsg` / `oncogene/tsg` / `unknown`), from `gene_roles.load_gene_roles`. AUTHORED

gene_dominated
GENE · BOOLEAN · 97%

Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction. BLOCK-LEVEL

gene_iqr
GENE · REAL â‰¥ 0 · 72%

interquartile range DERIVED

> *Defined on: genes with a NAT-TUM gene-level leg*

gene_lfc_NAT_TUM
GENE · REAL, SIGNED · 72%

in matched normal (NAT) Â· in tumour DERIVED

> *Defined on: genes with a NAT-TUM gene-level leg*

gene_net_repr
GENE · BOOLEAN · 100%

Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction. BLOCK-LEVEL

gene_repr_class
GENE · CATEGORICAL · 100%

class label DERIVED

AGO_ Â· 2

supporting â€" read these when the core raises a question

> **2** columns Â· coverage **95%â€"100%** (median 97%)
> 1× fraction 0â€"1 Â· 1× boolean

ago_loading
ARM · FRACTION 0â€"1 · 95%

AGO/RISC loading discount. AUTHORED

ago_loading_measured
ARM · BOOLEAN · 100%

Was `ago_loading` actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%). AUTHORED

> *Defined on: arms with a Manakov AGO-dominance measurement â€" â›" NOT loading(), which fabricates 1.0*

PIP_ Â· 2

supporting â€" read these when the core raises a question

> **2** columns Â· coverage **100%â€"100%** (median 100%)
> 1× constant Â· 1× fraction 0â€"1

pip_dense
EDGE · CONSTANT · 100%

âš CONSTANT by construction â€" the coupling readout is called with Ï€â‰¡1, so every edge has the same value. AUTHORED

pip_discovery
EDGE · FRACTION 0â€"1 · 100%

Posterior inclusion probability. `pip_dense` comes from the pi=1 coupling readout; `pip_discovery` from the evidence-pi readout. AUTHORED

RETENTION_ Â· 3

reference â€” context, provenance and rarely-quoted detail

3 columns Â· coverage **72%â€“100%** (median 100%)
1× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

retention_beta
EDGE | REAL â‰¥ 0 | 100%

Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

retention_reliable
EDGE | BOOLEAN | 100%

Is this edge's `retention` ratio trustworthy â€” i.e. does it have a real denominator? AUTHORED

retention_rho
EDGE | REAL, SIGNED | 72%

Ï_deconv / Ï_core â€” AUTHORED

OOF_ Â· 3

reference â€” context, provenance and rarely-quoted detail

Out-of-fold predictive correlation, arm-level vs family-level, and their difference (`oof_drho` = arm minus family). Negative = the arm-resolved fit predicts repression better.

3 columns Â· coverage **20%â€“20%** (median 20%) Â· **2** share the block description
3× signed âˆ’1â€¦1

oof_drho
FAMILY | SIGNED âˆ’1â€¦1 | 20%

Arm-level OOF rho MINUS family-level. AUTHORED

oof_rho_arm
FAMILY | SIGNED âˆ’1â€¦1 | 20%

Spearman Ï Â· per arm DERIVED

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€” 20.3% of edges*

oof_rho_fam
FAMILY | SIGNED âˆ’1â€¦1 | 20%

Spearman Ï Â· per seed family DERIVED

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€” 20.3% of edges*

OWN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **88%â€“88%** (median 88%)
1× real â‰¥ 0

own_specific_frac
ARM | REAL â‰¥ 0 | 88%

Fraction of the arm's variance that is target-specific. AUTHORED

GRANK_ Â· 3

reference â€” context, provenance and rarely-quoted detail

The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.

3 columns Â· coverage **63%â€“69%** (median 68%) Â· **3** share the block description
3× count

grank_HLY
ARM COUNT 63%

in the GTEx-healthy state DERIVED

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

grank_NAT
ARM COUNT 68%

in matched normal (NAT) DERIVED

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

grank_TUM
ARM COUNT 69%

in tumour DERIVED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

W_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

w_max
GENE REAL â‰¥ 0 100%

Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat. AUTHORED

COMPOSITION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× categorical

composition_class
EDGE CATEGORICAL 100%

Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block). AUTHORED

M_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

m_nnls
EDGE REAL â‰¥ 0 100%

Bagged-NNLS weight. Exactly 0 in 31.7% of families â€” this is what lets `identity` say a regulator contributes nothing, which `beta` structurally cannot. AUTHORED

PRIOR_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× constant

prior_pi
EDGE | CONSTANT | 100%

The Gibbs sampler's inclusion prior â€” **CONSTANT 0.05** across the card. AUTHORED

EDGE_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **73%â€“73%** (median 73%)
1× signed âˆ’1â€¦1

edge_rho_adj
EDGE | SIGNED Âˆ'1Â€¦1 | 73%

An edge-rung quantity carried onto this row. AUTHORED

> *Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)*

TERM_ Â· 3

reference â€” context, provenance and rarely-quoted detail

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms.
3 columns Â· coverage **68%â€“68%** (median 68%) Â· **3** share the block description
2× signed âˆ’1â€¦1 Â· 1× real, signed

term_ABUND
EDGE | SIGNED Âˆ'1Â€¦1 | 68%

unweighted abundance-sum reference DERIVED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

term_INTERACT
EDGE | SIGNED Âˆ'1Â€¦1 | 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms. BLOCK-LEVEL

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

term_WIRING
EDGE | REAL, SIGNED | 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms. BLOCK-LEVEL

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

MEAN_ Â· 2

reference â€” context, provenance and rarely-quoted detail

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises.
2 columns Â· coverage **90%â€“90%** (median 90%) Â· **2** share the block description
2× real, signed

mean_dGlobalRank
ARM | REAL, SIGNED | 90%

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises. BLOCK-LEVEL

> *Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)*

mean_own_shift
ARM | REAL, SIGNED | 90%

shift DERIVED

> *Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)*

SURROGATE_ Â· 2

reference â€" context, provenance and rarely-quoted detail

> Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for.
> **2** columns Â· coverage **3%â€"3%** (median 3%) Â· **2** share the block description
> 1× fraction 0â€"1 Â· 1× categorical

surrogate_corr
ARM | FRACTION 0â€"1 | 3%

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for. BLOCK-LEVEL

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€" 3.6% of edges*

surrogate_instrument
ARM | CATEGORICAL | 3%

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for. BLOCK-LEVEL

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€" 3.6% of edges*

ESUB_ Â· 8

reference â€" context, provenance and rarely-quoted detail

> PAM50 subtype-stratified coupling at the EDGE rung â€" one fit per subtype plus the pooled fit and an adjusted q.
> **8** columns Â· coverage **98%â€"100%** (median 99%) Â· **5** share the block description
> 6× signed âˆ'1â€¦1 Â· 1× categorical Â· 1× p-value

esub_best_sub
EDGE | CATEGORICAL | 100%

the best DERIVED

> *Defined on: edges with a per-PAM50 heterogeneity fit â€" âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_q_adj
EDGE | P-VALUE | 98%

BH-adjusted q for the best-subtype contrast â€" AUTHORED

esub_rho_Basal
EDGE | SIGNED Âˆ'1Â€¦1 | 99%

Spearman Ï DERIVED

> *Defined on: edges with a per-PAM50 heterogeneity fit â€" âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_Her2
EDGE | SIGNED Âˆ'1Â€¦1 | 98%

Spearman Ï DERIVED

> *Defined on: edges with a per-PAM50 heterogeneity fit â€" âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_LumA
EDGE | SIGNED Âˆ'1Â€¦1 | 99%

Spearman Ï DERIVED

> *Defined on: edges with a per-PAM50 heterogeneity fit â€" âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_LumB
EDGE | SIGNED Âˆ'1Â€¦1 | 99%

Spearman Ï DERIVED

> *Defined on: edges with a per-PAM50 heterogeneity fit â€" âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_best
EDGE | SIGNED Âˆ'1Â€¦1 | 100%

Coupling in the arm's BEST PAM50 subtype. AUTHORED

esub_rho_pool
EDGE | SIGNED Âˆ'1Â€¦1 | 100%

Coupling pooled across subtypes â€" the unselected reference for `esub_rho_best` (median âˆ'0.0135, indistinguishable from a random-subtype draw at âˆ'0.0123). AUTHORED

WIRING_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **68%â€“68%** (median 68%)
1× fraction 0â€“1

wiring_frac
EDGE | FRACTION 0Â€“1 | 68%

Fraction of the acquisition score attributable to WIRING rather than abundance. AUTHORED

Defined on: edges present in the progression panel (arm expressed + state measured)

KD_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· coverage **85%â€“85%** (median 85%)
1× percent Â· 1× real, signed

kd_affinity_pct
EDGE | PERCENT | 85%

Is this gene among the arm's strongest predicted targets â€” a WITHIN-ARM percentile, **higher = stronger target**. AUTHORED

kd_repression
EDGE | REAL, SIGNED | 85%

scanMiR RBNS binding affinity. AUTHORED

REALIZATION_ Â· 2

reference â€” context, provenance and rarely-quoted detail

Within-patient realization quantities â€” the score, its identity allocation, and who owns it.
2 columns Â· coverage **58%â€“69%** (median 63%) Â· **2** share the block description
1× fraction 0â€“1 Â· 1× real, signed

realization_identity
EDGE | FRACTION 0Â€“1 | 58%

Shapley identity DERIVED

Defined on: genes with a family-level realization fit

realization_score
EDGE | REAL, SIGNED | 69%

Within-patient realization quantities â€” the score, its identity allocation, and who owns it. BLOCK-LEVEL

Defined on: edges present in the progression panel (arm expressed + state measured)

CELL_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **20%â€“20%** (median 20%)
1× boolean

cell_arms_resolvable
FAMILY | BOOLEAN | 20%

Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells. AUTHORED

SD_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **20%â€“20%** (median 20%)
1× real â‰¥ 0

sd_arm
EDGE | REAL Â‰¥ 0 | 20%

Posterior SD of the arm-level Î². AUTHORED

ACQUISITION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

> **1** columns Â· coverage **59%â€“59%** (median 59%)
> 1× real, signed

acquisition_score
EDGE | REAL, SIGNED | 59%

Composite acquisition score across states. AUTHORED

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

Z_ Â· 1

reference â€” context, provenance and rarely-quoted detail

> **1** columns Â· coverage **20%â€“20%** (median 20%)
> 1× real â‰¥ 0

z_arm
EDGE | REAL Â‰¥ 0 | 20%

`beta_arm / sd_arm` from the WITHIN-CELL fit (`arm_rung.py`) â€” the arm's z inside its (gene, seed_family) cell only, defined on the 20.3% of edges in multi-arm cells. AUTHORED

HEUR_ Â· 1

reference â€” context, provenance and rarely-quoted detail

> **1** columns Â· coverage **100%â€“100%** (median 100%)
> 1× count

heur_gene_nreg
GENE | COUNT | 100%

Regulator count for this gene from the Â§6b-RETIRED heuristic lane. AUTHORED

FAMILY_ Â· 1

reference â€” context, provenance and rarely-quoted detail

> **1** columns Â· coverage **69%â€“69%** (median 69%)
> 1× signed âˆ’1â€¦1

family_rho_adj
FAMILY | SIGNED Âˆ'1Â€¦1 | 69%

Family-level properties carried onto this row (size, the arm's dose share of it, and its role). AUTHORED

CPTAC_ Â· 20

reference â€” context, provenance and rarely-quoted detail

> CPTAC protein channel.
>
> **20** columns Â· coverage **45%â€“79%** (median 65%) Â· **16** share the block description
> 12× signed âˆ'1â€¦1 Â· 4× real, signed Â· 2× constant Â· 2× categorical

cptac_prosp_comp_class_prot
EDGE | CATEGORICAL | 59%
Composition class of this edge's PROTEIN coupling (cell_intrinsic / partial / composition_explained) in the prospective cohort. AUTHORED

---

cptac_prosp_n
EDGE | CONSTANT | 79%
Patients in the prospective CPTAC cohort â€” **CONSTANT 101**. AUTHORED

---

cptac_prosp_ret_prot
EDGE | REAL, SIGNED | 59%
CPTAC prospective cohort (nâ‰ˆ101) Â· against protein DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_ret_rna
EDGE | REAL, SIGNED | 61%
CPTAC prospective cohort (nâ‰ˆ101) Â· against mRNA DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_disc
EDGE | SIGNED Âˆ’1Â€¦1 | 78%
CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against the discovery layer DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_disc_raw
EDGE | SIGNED Âˆ’1Â€¦1 | 78%
CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€” no confounder block DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_prot
EDGE | SIGNED Âˆ’1Â€¦1 | 78%
CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against protein DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_prot_raw
EDGE | SIGNED Âˆ’1Â€¦1 | 79%
UNADJUSTED protein coupling, prospective. AUTHORED

---

cptac_prosp_rho_rna
EDGE | SIGNED Âˆ’1Â€¦1 | 79%
CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against mRNA DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_rna_raw
EDGE | SIGNED Âˆ’1Â€¦1 | 79%
CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€” no confounder block DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_comp_class_prot
EDGE | CATEGORICAL | 45%
CPTAC TCGA-overlap cohort (n=105) Â· class label Â· against protein DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_n
EDGE | CONSTANT | 70%
Patients in the TCGA-overlap CPTAC cohort â€” **CONSTANT 105**. See the prospective twin. AUTHORED

---

cptac_t105_ret_prot
EDGE | REAL, SIGNED | 45%
CPTAC TCGA-overlap cohort (n=105) Â· against protein DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_ret_rna
EDGE | REAL, SIGNED | 49%
CPTAC TCGA-overlap cohort (n=105) Â· against mRNA DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_disc
EDGE | SIGNED Âˆ’1Â€¦1 | 64%
CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against the discovery layer DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_disc_raw
EDGE | SIGNED Âˆ’1Â€¦1 | 65%
CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€” no confounder block DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_prot
EDGE | SIGNED Âˆ’1Â€¦1 | 64%
CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against protein DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_prot_raw
EDGE | SIGNED Âˆ’1Â€¦1 | 65%
CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against protein Â· UNADJUSTED â€” no confounder block DERIVED
> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_t105_rho_rna`
EDGE | SIGNED Â˘'1Â€¦1 | 64%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against mRNA DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_t105_rho_rna_raw`
EDGE | SIGNED Â˘'1Â€¦1 | 65%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€" no confounder block DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

HLY_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **61%â€"61%** (median 61%)
1× boolean

`hly_leg_concordant`
EDGE | BOOLEAN | 61%

Do the two HLYâ†'TUM legs (`_QN` and `_raw`) at least AGREE ON SIGN. **Concordant 76.1%, DISCORDANT 23.9%** of the 3,433 edges carrying both. AUTHORED

IS_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **58%â€"58%** (median 58%)
1× boolean

`is_realization_owner`
EDGE | BOOLEAN | 58%

A boolean flag; read the suffix. AUTHORED

> *Defined on: genes with a family-level realization fit*

realization/gene_card.tsv

gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance â€” `agg_of` carries the meaning instead.

135 columns **1,549** rows **29** blocks median coverage **83%**

Contents â€” 29 blocks, most important first

N_ Â· 7

core â€” the columns findings rest on

7 columns Â· coverage **91%â€“100%** (median 100%)
7× count

n_arms GENE COUNT AGG ARM 100%	Arms in the LEARNED MODEL's design â€” `X.shape[1]` from `assemble_gene`, carried via `gene_atlas`. **This is the width the Gibbs actually fit** (median 2, max 90). AUTHORED
n_cell_intrinsic GENE COUNT 100%	FAMILIES whose coupling SURVIVES the composition block (retention â‰¥ 0.7). AUTHORED
n_composition_explained GENE COUNT 100%	FAMILIES whose coupling does NOT survive the composition block (retention < 0.4). AUTHORED
n_fam GENE COUNT AGG FAMILY 100%	How many seed families regulate this gene. AUTHORED
n_hallmark_sets GENE COUNT 91%	How many MSigDB Hallmark programs contain this gene (1â€“10, median 2). Membership only â€” it says nothing about regulation. AUTHORED
n_identified GENE COUNT 100%	Families reaching |z| > 2 on the UNCALIBRATED SD. **Zero for 691 genes (44.6%)** â€” meaning the model cannot distinguish ANY of that gene's families from zero. AUTHORED

n_pip_disc_gt50
GENE COUNT 100%

Count of the gene's families with `pip_discovery > 0.5` under the evidence-Ï€ readout. AUTHORED

(BARE) Â· 2

core â€” the columns findings rest on

> **2** columns Â· coverage **100%â€“100%** (median 100%)
> 1× fraction 0â€“1 Â· 1× identifier

concentration
GENE FRACTION 0Â€“1 100%

`beta.max() / beta.sum()` â€” the TOP FAMILY's share of the gene's total Î²; median **0.954**, i.e. most genes are dominated by one family. AUTHORED

---

gene
KEY IDENTIFIER 100%

KEY â€” HGNC gene symbol. AUTHORED

COMP_ Â· 16

core â€” the columns findings rest on

> Cell-composition drivers of this gene's coupling (`compartment_*`): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
>
> **16** columns Â· coverage **59%â€“90%** (median 73%) Â· **7** share the block description
> 6× categorical Â· 6× signed âˆ’1â€¦1 Â· 3× real, signed Â· 1× real â‰¥ 0

comp_cptac_prot_driver
GENE CATEGORICAL 59%

CPTAC Â· against protein DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_driver_ret
GENE REAL, SIGNED 59%

`1 âˆ’ comp_cptac_prot_driver_share`, exactly. Median +0.398 raw, **+0.469** gated to |Ï_raw| â‰¥ 0.10 â€” quote the gated value. See the mRNA twin. AUTHORED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_driver_share
GENE REAL, SIGNED 59%

Protein-layer twin of `comp_tcga_mrna_driver_share`. Same exact complementarity with `_driver_ret` (8.9e-16 on 918 rows). AUTHORED

---

comp_cptac_prot_rho_raw
GENE SIGNED Â−1Â€¦1 73%

Unadjusted aggregate PROTEIN coupling â€” denominator of `comp_cptac_prot_driver_ret`. AUTHORED

---

comp_cptac_prot_top_budget
GENE CATEGORICAL 73%

CPTAC Â· against protein Â· the top-ranked Â· pressure budget DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_budget_r
GENE SIGNED Â−1Â€¦1 73%

CPTAC Â· against protein Â· the top-ranked Â· pressure budget Â· correlation DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_target
GENE CATEGORICAL 73%

CPTAC Â· against protein Â· the top-ranked DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_target_r
GENE SIGNED Â−1Â€¦1 73%

CPTAC Â· against protein Â· the top-ranked Â· correlation DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_tcga_mrna_driver
GENE CATEGORICAL 66%

WHICH compartment drives the gene's mRNA coupling â€” the one whose removal moves it most (`CAFs` 279 Â· `purity` 174 Â· `prolif` 148 Â· `Endothelial` 122 Â· `Myeloid` 109). AUTHORED

---

comp_tcga_mrna_driver_ret
GENE | REAL, SIGNED | 66%

`1 âˆ’ comp_tcga_mrna_driver_share`, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED

---

comp_tcga_mrna_driver_share
GENE | REAL Â‰¥ 0 | 66%

Share of the gene's mRNA coupling attributable to the single compartment that most changes it â€” **axiom 8's composition GATE**, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED

---

comp_tcga_mrna_rho_raw
GENE | SIGNED Âˆ'1Â€¦1 | 90%

The gene's UNADJUSTED aggregate mRNA coupling â€” the DENOMINATOR of `comp_tcga_mrna_driver_ret`. AUTHORED

---

comp_tcga_mrna_top_budget
GENE | CATEGORICAL | 90%

The compartment the gene's miRNA BUDGET loads on most â€” a property of the REGULATORS. AUTHORED

---

comp_tcga_mrna_top_budget_r
GENE | SIGNED Âˆ'1Â€¦1 | 90%

TCGA Â· the top-ranked Â· pressure budget Â· correlation DERIVED

> *Defined on: genes/families with a TCGA-mRNA compartment driver â€” âš MH-201: these axes predict the SIGN of Î²'s gain*

---

comp_tcga_mrna_top_target
GENE | CATEGORICAL | 90%

The compartment the TARGET GENE's expression loads on most. AUTHORED

---

comp_tcga_mrna_top_target_r
GENE | SIGNED Âˆ'1Â€¦1 | 90%

TCGA Â· the top-ranked Â· correlation DERIVED

> *Defined on: genes/families with a TCGA-mRNA compartment driver â€” âš MH-201: these axes predict the SIGN of Î²'s gain*

CTX_ Â· 16

core â€” the columns findings rest on

> Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
>
> **16** columns Â· coverage **45%â€“100%** (median 90%) Â· **13** share the block description
> 4× signed âˆ'1â€¦1 Â· 3× real â‰¥ 0 Â· 3× real, signed Â· 3× count Â· 1× categorical Â· 1× fraction 0â€“1 Â· 1× boolean

ctx_apriori_class
GENE | CATEGORICAL | 100%

A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED). AUTHORED

---

ctx_ceiling
GENE | SIGNED Âˆ'1Â€¦1 | 98%

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. BLOCK-LEVEL

---

ctx_collapse
GENE | REAL Â‰¥ 0 | 90%

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. BLOCK-LEVEL

---

ctx_d_collin
GENE | SIGNED Âˆ'1Â€¦1 | 45%

change DERIVED

> *Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)*

---

ctx_d_fam
GENE | REAL, SIGNED | 87%

change Â· per seed family DERIVED

---

ctx_dose_max
GENE | REAL Â‰¥ 0 | 90%

dose Â· maximum DERIVED

---

ctx_dose_med
GENE | REAL Â‰¥ 0 | 90%

dose Â· median DERIVED

ctx_frac_abund
GENE | FRACTION 0â€“1 | 90%

fraction Â· unweighted abundance-sum reference DERIVED

ctx_gap_core
GENE | SIGNED âˆ'1â€¦1 | 87%

Real-minus-decoy coupling gap for this gene. Pooled it is ~-0.012, but the per-gene IQR STRADDLES ZERO â€" this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it. AUTHORED

ctx_gap_d_dose
GENE | REAL, SIGNED | 87%

gap Â· change Â· dose DERIVED

ctx_gap_deconv
GENE | SIGNED âˆ'1â€¦1 | 87%

gap Â· composition-adjusted (deconvolution block) DERIVED

ctx_gap_retention
GENE | REAL, SIGNED | 83%

gap Â· retention DERIVED

ctx_measurable
GENE | BOOLEAN | 98%

Is the gene's coupling measurable at all given its design (positive OOF ceiling). False for 877/1549 genes. AUTHORED

ctx_n_abund
GENE | COUNT | 90%

count Â· unweighted abundance-sum reference DERIVED

ctx_n_fam_abund
GENE | COUNT | 90%

count Â· per seed family Â· unweighted abundance-sum reference DERIVED

ctx_n_fam_multi
GENE | COUNT | 90%

count Â· per seed family DERIVED

TOP_ Â· 8

supporting â€" read these when the core raises a question

> The gene's leading regulator and its value, by magnitude or by identity.
>
> **8** columns Â· coverage **78%â€“100%** (median 89%) Â· **3** share the block description
>
> 3× identifier Â· 2× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× signed âˆ'1â€¦1 Â· 1× count

top_beta
GENE | REAL â‰¥ 0
AGG FAMILY | 100%

Î² DERIVED

top_beta_frac
GENE | FRACTION 0â€“1
AGG FAMILY | 100%

Î² Â· fraction DERIVED

top_family_identity
GENE | IDENTIFIER
AGG FAMILY | 87%

The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything â€" an honest 'undefined', which the magnitude answer cannot express. AUTHORED

top_family_magnitude
GENE | IDENTIFIER
AGG FAMILY | 100%

The family with the largest beta. AUTHORED

top_gaining_arms
GENE | IDENTIFIER | 91%

arms DERIVED

top_identity
GENE | REAL Â‰¥ 0
AGG FAMILY | 87%

The largest `identity` share among the gene's families. AUTHORED

top_identity_gated
GENE | SIGNED Âˆ'1Â€¦1 | 78%

`top_identity` recomputed over `identity_reliable` edges only â€" **max 1.0000, 0 genes above 1** (vs +740.007 ungated). Defined on 1,203 genes. AUTHORED

top_identity_n_reliable
GENE | COUNT | 78%

How many of the gene's families survive the identity gate â€" the DENOMINATOR behind `top_identity_gated`. **Median 2.** AUTHORED

REALIZED_ Â· 5

supporting â€" read these when the core raises a question

> Realized (as opposed to potential) coupling for the gene â€" the raw and adjusted rho, its retention, and how many regulators it was computed over.
>
> **5** columns Â· coverage **63%â€"82%** (median 81%) Â· **4** share the block description
> 2× signed âˆ'1â€¦1 Â· 1× boolean Â· 1× count Â· 1× real, signed

realized_composition_explained
GENE | BOOLEAN | 82%

Realized (as opposed to potential) coupling for the gene â€" the raw and adjusted rho, its retention, and how many regulators it was computed over. BLOCK-LEVEL

realized_n_reg
GENE | COUNT | 82%

count DERIVED

realized_retention
GENE | REAL, SIGNED | 63%

Realized-coupling retention, Ï_adjusted / Ï_raw. AUTHORED

realized_rho_adj
GENE | SIGNED Âˆ'1Â€¦1 | 81%

Spearman Ï Â· composition-adjusted DERIVED

realized_rho_raw
GENE | SIGNED Âˆ'1Â€¦1 | 81%

Spearman Ï Â· UNADJUSTED â€" no confounder block DERIVED

W_ Â· 1

supporting â€" read these when the core raises a question

> **1** columns Â· coverage **100%â€"100%** (median 100%)
> 1× real â‰¥ 0

w_max
GENE | REAL Â‰¥ 0 | 100%

The MAXIMUM curated evidence weight `w` over the gene's regulators (`gene_atlas`: `nanmax(w)`), where `w` comes from the same PMID-deduped ledger that feeds `lit_` and `fame_`. AUTHORED

LIT_ Â· 7

supporting â€” read these when the core raises a question

â›"â›" **A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH â€" read the name with suspicion.** `eval/lit_ground_truth.py` defines the 'canonical' family of a gene as the **argmax of DISTINCT PubMed IDs** carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded).

7 columns Â· coverage **21%â€"21%** (median 21%) Â· **3** share the block description
4× count Â· 2× boolean Â· 1× identifier

lit_agrees_identity
GENE BOOLEAN 21%

Does `identity` name the same family as the **most-published** one for this gene â€" **True on 100 of 323 (31%).** AUTHORED

lit_agrees_magnitude
GENE BOOLEAN 21%

â›"â›" **A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH â€" read the name with suspicion.** `eval/lit_ground_truth.py` defines the 'canonical' family of a gene as the **argmax of DISTINCT PubMed IDs** carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded). BLOCK-LEVEL

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

lit_family
GENE IDENTIFIER 21%

The most-published seed family for this gene by distinct low-throughput functional PMIDs. AUTHORED

lit_margin
GENE COUNT 21%

Gap between the top and second literature family, in PMIDs â€" **median 1.00**, i.e. the literature's 'winner' is usually decided by a SINGLE paper. â‡' **gate on this before reading `lit_agrees_*`**: a 1-PMID margin is a coin flip dressed as ground truth. AUTHORED

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

lit_n_lit_families
GENE COUNT 21%

count DERIVED

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

lit_n_pmid
GENE COUNT 21%

Distinct low-throughput functional PMIDs behind `lit_family`. Median **3** â€" the 'ground truth' rests on a handful of papers per gene. AUTHORED

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

lit_tier
GENE COUNT 21%

â›"â›" **A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH â€" read the name with suspicion.** `eval/lit_ground_truth.py` defines the 'canonical' family of a gene as the **argmax of DISTINCT PubMed IDs** carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded). BLOCK-LEVEL

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

CPTAC_ Â· 24

reference â€" context, provenance and rarely-quoted detail

CPTAC protein channel.

24 columns Â· coverage **42%â€"89%** (median 66%) Â· **18** share the block description
18× signed âˆ'1â€¦1 Â· 2× boolean Â· 2× real, signed Â· 2× count

cptac_prosp_abund_rho_disc
GENE SIGNED Âˆ'1â€¦1
AGG ARM 72%

CPTAC prospective cohort (nâ‰ˆ101) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against the discovery layer DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_prosp_abund_rho_prot
GENE SIGNED Âˆ'1â€¦1
AGG ARM 73%

CPTAC prospective cohort (nâ‰ˆ101) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against protein DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_abund_rho_rna`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `74%`

CPTAC prospective cohort (nâ‰ˆ101) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against mRNA `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_beats_abund_prot`
`GENE` `BOOLEAN` `73%`

Does the learned Î²-weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort? `AUTHORED`

`cptac_prosp_agg_ret_prot`
`GENE` `REAL, SIGNED` `59%`

`rho_prot(composition-adjusted) / rho_prot_raw` â€” the ADJUSTMENT sense of retention, on the prospective cohort. âœ… **The denominator is ALREADY GATED** at `RHO_GATE` in `cptac_card.py:322`, so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with |raw| < 0.02). `AUTHORED`

`cptac_prosp_agg_rho_disc`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `72%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_disc_raw`
`GENE` `SIGNED Â^'1Â€¦1` `72%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€” no confounder block `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_prot`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `73%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_prot_raw`
`GENE` `SIGNED Â^'1Â€¦1` `73%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein Â· UNADJUSTED â€” no confounder block `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_rna`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `74%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_rna_raw`
`GENE` `SIGNED Â^'1Â€¦1` `74%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€” no confounder block `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_n_arms`
`GENE` `COUNT` `89%`

Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61). `AUTHORED`

`cptac_t105_abund_rho_disc`
`GENE` `SIGNED Â^'1Â€¦1` `60%`

CPTAC TCGA-overlap cohort (n=105) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against the discovery layer `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_t105_abund_rho_prot`
`GENE` `SIGNED Â^'1Â€¦1` `60%`

CPTAC TCGA-overlap cohort (n=105) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against protein `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_t105_abund_rho_rna`
`GENE` `SIGNED Â^'1Â€¦1` `60%`

CPTAC TCGA-overlap cohort (n=105) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against mRNA `DERIVED`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_t105_agg_beats_abund_prot`
`GENE` `BOOLEAN` `60%`

As the prospective twin, on the TCGA-105 overlap cohort. `AUTHORED`

`cptac_t105_agg_ret_prot`
`GENE` `REAL, SIGNED` `42%`

As the prospective twin, TCGA-105. `AUTHORED`

---

cptac_t105_agg_rho_disc
GENE | SIGNED Â˘'1â€¦1 | AGG ARM | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_t105_agg_rho_disc_raw
GENE | SIGNED Â˘'1â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€" no confounder block DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_t105_agg_rho_prot
GENE | SIGNED Â˘'1â€¦1 | AGG ARM | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_t105_agg_rho_prot_raw
GENE | SIGNED Â˘'1â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein Â· UNADJUSTED â€" no confounder block DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_t105_agg_rho_rna
GENE | SIGNED Â˘'1â€¦1 | AGG ARM | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_t105_agg_rho_rna_raw
GENE | SIGNED Â˘'1â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€" no confounder block DERIVED

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_t105_n_arms
GENE | COUNT | 85%

Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. AUTHORED

> *Defined on: genes/edges present in the CPTAC cohort*

DOMINANT_ Â· 4

reference â€" context, provenance and rarely-quoted detail

> The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to `regulatory_handoff`.
>
> **4** columns Â· coverage **77%â€"83%** (median 82%) Â· **3** share the block description
> 3× identifier Â· 1× categorical

dominant_HLY
GENE | IDENTIFIER | 77%

in the GTEx-healthy state DERIVED

dominant_NAT
GENE | IDENTIFIER | 81%

in matched normal (NAT) DERIVED

dominant_TUM
GENE | IDENTIFIER | 82%

The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next). AUTHORED

dominant_edge_shift_class
GENE | CATEGORICAL | 83%

per edge Â· shift Â· class label DERIVED

FRAC_ Â· 3

reference â€" context, provenance and rarely-quoted detail

> A fraction â€" read the suffix for the numerator and `domain` for the denominator's rows.
>
> **3** columns Â· coverage **60%â€"100%** (median 82%) Â· **3** share the block description
> 3× fraction 0â€"1

frac_edges_realized
GENE | FRACTION 0â€“1 | 82%

A fraction â€” read the suffix for the numerator and `domain` for the denominator's rows. BLOCK-LEVEL

frac_gain_true_acquired
GENE | FRACTION 0â€“1 | 60%

A fraction â€” read the suffix for the numerator and `domain` for the denominator's rows. BLOCK-LEVEL

> *Defined on: genes with a NAT-TUM gene-level leg*

frac_identified
GENE | FRACTION 0â€“1 | 100%

A fraction â€” read the suffix for the numerator and `domain` for the denominator's rows. BLOCK-LEVEL

STATIC_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **87%â€“87%** (median 87%)
1× identifier

static_owner_family
GENE | IDENTIFIER | 87%

The static (dose-based) owner of the gene, for comparison against the realization owner. AUTHORED

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

TOTAL_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

total_pressure
GENE | REAL Â‰¥ 0 | 100%

Sum over the gene's regulators. AUTHORED

OWNER_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **41%â€“41%** (median 41%)
1× boolean

owner_agrees
GENE | BOOLEAN | 41%

Whether the realization owner and the static owner agree. Defined only where both exist. AUTHORED

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

MEDIAN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

median_retention
GENE | REAL Â‰¥ 0 | 100%

A per-row median of a lower-rung quantity; check `agg_of`. AUTHORED

ARMRES_ Â· 7

reference â€” context, provenance and rarely-quoted detail

â IS THIS GENE MODELABLE AT ARM RESOLUTION â€” the gene-level roll-up of arm identifiability (`card_ladders.gene_arm_resolution` , 2026-08-19).

7 columns Â· coverage **18%â€“18%** (median 18%) Â· **4** share the block description
2× signed âˆ’1â€¦1 Â· 2× count Â· 1× categorical Â· 1× fraction 0â€“1 Â· 1× real â‰¥ 0

armres_best_drho
GENE · SIGNED Âˆ'1Â€¦1 · 18%
the best DERIVED

armres_class
GENE · CATEGORICAL · 18%
`arm_modelable` (every multi-arm cell resolvable, 68 genes) Â· `partial` (37) Â· `family_only` (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined. AUTHORED

armres_frac_resolvable
GENE · FRACTION 0Â€"1 · 18%
Fraction of this gene's multi-arm cells where the arms are separable. AUTHORED

armres_med_drho
GENE · SIGNED Âˆ'1Â€¦1 · 18%
Median `oof_drho` over the gene's multi-arm cells â€” negative means arm resolution predicts repression better out of fold. AUTHORED

armres_med_sep_z
GENE · REAL Â‰¥ 0 · 18%
median Â· z-score DERIVED

armres_n_arms_split
GENE · COUNT · 18%
count Â· arms DERIVED

armres_n_multi_cells
GENE · COUNT · 18%
count DERIVED

IDENTITY_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **87%â€“87%** (median 87%)
1× boolean

identity_eq_magnitude
GENE · BOOLEAN · AGG FAMILY · 87%
Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at n_fam==1 by construction; 76% at n_fam>=2. Read it only on multi-family genes. AUTHORED

HEUR_ Â· 5

reference â€” context, provenance and rarely-quoted detail

5 columns Â· coverage **89%â€“91%** (median 91%)
2× signed âˆ’1â€¦1 Â· 1× count Â· 1× boolean Â· 1× categorical

heur_delta_tumor_nat
GENE · SIGNED Âˆ'1Â€¦1 · 89%
Tumour-minus-NAT change in the heuristic pressure correlation. AUTHORED

heur_n_regulators
GENE · COUNT · 91%
âš âš **A DIFFERENT PIPELINE'S COUNT â€” not the model's.** It is `edges.groupby('gene')['miRNA'].nunique()` inside `analyses/misc/mirna_comovement.py` , i.e. AUTHORED

heur_net_repressed_tumor
GENE BOOLEAN 91%

Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes. AUTHORED

heur_repression_class
GENE CATEGORICAL 91%

Cross-state repression class from the RETIRED heuristic lane (never_repressed / lost_repression / gained_repression / constitutive_repressed). 867 of 1,409 genes are `never_repressed`. AUTHORED

heur_rho_pressure_tumor
GENE SIGNED Âˆ'1Â€¦1 91%

Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour. AUTHORED

GREAL_ Â· 10

reference â€" context, provenance and rarely-quoted detail

Realization ladder for the GENE â€" the mirror of `real_`: how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). âœ... Cross-rung control: the gene-side totals reproduce the arm-side exactly.

10 columns Â· coverage **29%â€"83%** (median 83%) Â· **8** share the block description
7× count Â· 2× signed âˆ'1â€¦1 Â· 1× fraction 0â€"1

greal_best_coupling
GENE SIGNED Âˆ'1Â€¦1 83%

the best DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_frac_c10
GENE FRACTION 0Â€"1 29%

Gene-rung twin of `real_frac_c10` â€" defined on 446 genes against `greal_n_c10`'s 1,289; gated-out genes have a median `greal_n_scored` of **1**. Exact on 100%. AUTHORED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_med_coupling
GENE SIGNED Âˆ'1Â€¦1 83%

median DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_c05
GENE COUNT 83%

count Â· at the âˆ'0.05 threshold DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_c10
GENE COUNT 83%

count Â· at the âˆ'0.10 threshold DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_c15
GENE COUNT 83%

count Â· at the âˆ'0.15 threshold DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_c20
GENE COUNT 83%

count Â· at the âˆ'0.20 threshold DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_c30
GENE COUNT 83%

count Â· at the âˆ'0.30 threshold DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_scored
GENE COUNT 83%

Genes/edges scored for realized coupling (median 2, max 59) â€" the DENOMINATOR of every `greal_frac_*`. âœ... The `greal_n_c05 â€¦ n_c30` ladder is properly nested: **0 monotonicity violations and 0 rows where a threshold count exceeds `n_scored`** (verified). AUTHORED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_sig

GENE | COUNT | 83%

count DERIVED

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

ACQUIRED_ Â· 2

reference â€" context, provenance and rarely-quoted detail

> Acquisition of regulatory dose relative to a healthy reference (GTEx or NAT).
>
> **2** columns Â· coverage **91%â€"91%** (median 91%) Â· **2** share the block description
>
> 2× real, signed

acquired_vs_gtex

GENE | REAL, SIGNED | 91%

GTEx DERIVED

---

acquired_vs_nat

GENE | REAL, SIGNED | 91%

in matched normal (NAT) DERIVED

PRESSURE_ Â· 1

reference â€" context, provenance and rarely-quoted detail

> **1** columns Â· coverage **91%â€"91%** (median 91%)
>
> 1× real â‰¥ 0

pressure_tumor

GENE | REAL Â‰¥ 0 | 91%

Aggregate incoming pressure on the gene in the named state. AUTHORED

REALIZATION_ Â· 1

reference â€" context, provenance and rarely-quoted detail

> **1** columns Â· coverage **42%â€"42%** (median 42%)
>
> 1× identifier

realization_owner

GENE | IDENTIFIER | 42%

Within-patient realization quantities â€" the score, its identity allocation, and who owns it. AUTHORED

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

TCGA_ Â· 4

reference â€" context, provenance and rarely-quoted detail

> The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.
>
> **4** columns Â· coverage **89%â€"89%** (median 89%) Â· **3** share the block description
>
> 3× signed âˆ'1â€¦1 Â· 1× count

tcga_abund_rho_rna

GENE | SIGNED Âˆ'1Â€¦1 | AGG ARM | 89%

unweighted abundance-sum reference Â· Spearman Ï Â· against mRNA DERIVED

---

tcga_agg_rho_rna
GENE | SIGNED Â^'1â€¦1 | AGG ARM | 89%

Gene-level Î²-weighted aggregate coupling to mRNA in TCGA (median **â^'0.05**) against `tcga_abund_rho_rna`'s **â^'0.02** â€" the abundance-sum reference. AUTHORED

tcga_agg_rho_rna_raw
GENE | SIGNED Â^'1â€¦1 | 89%

Î²-weighted aggregate Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€" no confounder block DERIVED

tcga_n_arms
GENE | COUNT | 89%

count Â· arms DERIVED

DISC_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· coverage **3%â€"100%** (median 51%)
1× categorical Â· 1× count

disc_gold_families
GENE | CATEGORICAL | 3%

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

disc_n_gold_edges
GENE | COUNT | 100%

How many of the 157 discovery-queue edges name this gene (max 4). AUTHORED

MAX_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€"100%** (median 100%)
1× fraction 0â€"1

max_beta_frac_sd
GENE | FRACTION 0â€"1 | 100%

Maximum over the row's constituent units. AUTHORED

REGULATORY_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€"100%** (median 100%)
1× boolean

regulatory_handoff
GENE | BOOLEAN | 100%

Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour. AUTHORED

MEAN_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **82%â€"82%** (median 82%)
1× real, signed

mean_dTarget
GENE | REAL, SIGNED | 82%

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises. AUTHORED

CONCENTRATION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **53%â€“53%** (median 53%)
1× fraction 0â€“1

concentration_adj
GENE | FRACTION 0Â€“1 | 53%

`(concentration âˆ’ 1/n_fam) / (1 âˆ’ 1/n_fam)` âˆˆ [0,1] â€” how concentrated the gene is RELATIVE to the most diffuse its own design allows. AUTHORED

FAM_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **81%â€“81%** (median 81%)
1× signed âˆ’1â€¦1

fam_pooled_rho_adj
GENE | SIGNED Âˆ’1Â€¦1 | 81%

Seed-family structure the arm sits in â€” membership, affinity and dominance within it. AUTHORED gene_family_card.tsv

gene_family

One row per one (seed family, gene) pair. Keyed [gene, family] â€” it cannot express a property of the family itself. That is the seed_family card's job.

61 columns **5,117** rows **12** blocks median coverage **97%**

Contents â€” 12 blocks, most important first

(BARE) Â· 8

core â€” the columns findings rest on

8 columns Â· coverage **95%â€“100%** (median 100%)
3× identifier Â· 3× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

arms FAMILY IDENTIFIER 100%	The arms collapsed into this (gene, family) cell, **semicolon-separated** (`hsa-miR-17-5p;hsa-miR-20a-5p`). AUTHORED
beta FAMILY REAL â‰¥ 0 100%	FAMILY-rung coupling coefficient â€” `readouts.run(level='family')` , the same-seed collapse APPLIED. AUTHORED
family KEY IDENTIFIER 100%	KEY â€” seed family of the regulator, as used by the family-rung fit. AUTHORED
gene KEY IDENTIFIER 100%	KEY â€” HGNC gene symbol. AUTHORED
identified FAMILY BOOLEAN 100%	`|z| > 2` on the UNCALIBRATED SD â€” **27.1%** of cells. `cal_identified` is the honest version. AUTHORED
identity FAMILY REAL, SIGNED 95%	Shapley/LMG identity at the FAMILY rung â€” same estimator as the edge card's `identity` , different unit. AUTHORED
retention FAMILY REAL â‰¥ 0 100%	`beta_deconv / beta_core` at the family grain â€” the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense. AUTHORED
z FAMILY REAL â‰¥ 0 100%	`beta / beta_sd` for the family cell, on the **UNCALIBRATED** posterior SD. AUTHORED

N_ Â· 3

supporting â€” read these when the core raises a question

```
3 columns  Â·  coverage 100%â€“100% (median 100%)
2× count Â· 1× constant
```

n_arms `FAMILY` `COUNT` `100%`	How many arms of THIS family sit in THIS gene's design. âœ… FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's `n_arms` exactly â€” a clean cross-rung consistency check. `AUTHORED`
n_fam `GENE` `COUNT` `100%`	How many families compete at this gene. beta is mathematically INERT at 1. `AUTHORED`
n_samples `GENE` `CONSTANT` `100%`	Samples the family cell was fit on â€” **CONSTANT 1,040** across the card. `AUTHORED`

CTX_ Â· 16

supporting â€” read these when the core raises a question

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

```
16 columns  Â·  coverage 81%â€“100% (median 96%)  Â·  16 share the block description
4× signed âˆ’1â€¦1 Â· 3× real â‰¥ 0 Â· 3× real, signed Â· 3× count Â· 1× categorical Â· 1× fraction 0â€“1 Â· 1× boolean
```

ctx_apriori_class `GENE` `CATEGORICAL` `100%`	class label `DERIVED`
ctx_ceiling `GENE` `SIGNED âˆ’1â€¦1` `99%`	Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. `BLOCK-LEVEL`
ctx_collapse `GENE` `REAL â‰¥ 0` `96%`	Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. `BLOCK-LEVEL`
ctx_d_collin `GENE` `SIGNED âˆ’1â€¦1` `81%`	change `DERIVED` *Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)*
ctx_d_fam `GENE` `REAL, SIGNED` `95%`	change Â· per seed family `DERIVED`
ctx_dose_max `GENE` `REAL â‰¥ 0` `96%`	dose Â· maximum `DERIVED`
ctx_dose_med `GENE` `REAL â‰¥ 0` `96%`	dose Â· median `DERIVED`
ctx_frac_abund `GENE` `FRACTION 0â€“1` `96%`	fraction Â· unweighted abundance-sum reference `DERIVED`
ctx_gap_core `GENE` `SIGNED âˆ’1â€¦1` `95%`	gap Â· core confounder block `DERIVED`
ctx_gap_d_dose `GENE` `REAL, SIGNED` `95%`	gap Â· change Â· dose `DERIVED`

ctx_gap_deconv
GENE · SIGNED Â˜’1Â€¦1 · 95%
gap Â· composition-adjusted (deconvolution block) DERIVED

ctx_gap_retention
GENE · REAL, SIGNED · 92%
gap Â· retention DERIVED

ctx_measurable
GENE · BOOLEAN · 99%
Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. BLOCK-LEVEL

ctx_n_abund
GENE · COUNT · 96%
count Â· unweighted abundance-sum reference DERIVED

ctx_n_fam_abund
GENE · COUNT · 96%
count Â· per seed family Â· unweighted abundance-sum reference DERIVED

ctx_n_fam_multi
GENE · COUNT · 96%
count Â· per seed family DERIVED

BETA_ Â· 7

supporting â€” read these when the core raises a question

> Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta.
>
> **7** columns Â· coverage **100%â€“100%** (median 100%) Â· **7** share the block description
> 4× fraction 0â€“1 Â· 3× real â‰¥ 0

beta_deconv
FAMILY · REAL Â‰¥ 0 · 100%
composition-adjusted (deconvolution block) DERIVED

beta_frac
FAMILY · FRACTION 0Â€“1 · 100%
fraction DERIVED

beta_frac_abs
FAMILY · FRACTION 0Â€“1 · 100%
fraction DERIVED

beta_frac_deconv
FAMILY · FRACTION 0Â€“1 · 100%
fraction Â· composition-adjusted (deconvolution block) DERIVED

beta_frac_sd
FAMILY · FRACTION 0Â€“1 · 100%
fraction Â· standard deviation DERIVED

beta_sd
FAMILY · REAL Â‰¥ 0 · 100%
standard deviation DERIVED

beta_sd_deconv
FAMILY · REAL Â‰¥ 0 · 100%
standard deviation Â· composition-adjusted (deconvolution block) DERIVED

Cell-composition drivers of this gene's coupling (`compartment_*`): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.

16 columns Â· coverage **66%â€“97%** (median 81%) Â· **7** share the block description
6× categorical Â· 6× signed â^'1â€¦1 Â· 3× real, signed Â· 1× real â‰¥ 0

comp_cptac_prot_driver
GENE | CATEGORICAL | 66%

CPTAC Â· against protein DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_driver_ret
GENE | REAL, SIGNED | 66%

`1 â^' comp_cptac_prot_driver_share`, exactly. Median +0.398 raw, **+0.469** gated to |Ï_raw| â‰¥ 0.10 â€” quote the gated value. See the mRNA twin. AUTHORED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_driver_share
GENE | REAL, SIGNED | 66%

Protein-layer twin of `comp_tcga_mrna_driver_share`. Same exact complementarity with `_driver_ret` (8.9e-16 on 918 rows). AUTHORED

comp_cptac_prot_rho_raw
GENE | SIGNED Â^'1Â€¦1 | 81%

Unadjusted aggregate PROTEIN coupling â€” denominator of `comp_cptac_prot_driver_ret`. AUTHORED

comp_cptac_prot_top_budget
GENE | CATEGORICAL | 81%

CPTAC Â· against protein Â· the top-ranked Â· pressure budget DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_budget_r
GENE | SIGNED Â^'1Â€¦1 | 81%

CPTAC Â· against protein Â· the top-ranked Â· pressure budget Â· correlation DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_target
GENE | CATEGORICAL | 81%

CPTAC Â· against protein Â· the top-ranked DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_target_r
GENE | SIGNED Â^'1Â€¦1 | 81%

CPTAC Â· against protein Â· the top-ranked Â· correlation DERIVED

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_tcga_mrna_driver
GENE | CATEGORICAL | 81%

WHICH compartment drives the gene's mRNA coupling â€” the one whose removal moves it most (`CAFs` 279 Â· `purity` 174 Â· `prolif` 148 Â· `Endothelial` 122 Â· `Myeloid` 109). AUTHORED

comp_tcga_mrna_driver_ret
GENE | REAL, SIGNED | 81%

`1 â^' comp_tcga_mrna_driver_share`, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED

comp_tcga_mrna_driver_share
GENE | REAL Â‰¥ 0 | 81%

Share of the gene's mRNA coupling attributable to the single compartment that most changes it â€” **axiom 8's composition GATE**, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED

comp_tcga_mrna_rho_raw
GENE | SIGNED Â^'1Â€¦1 | 97%

The gene's UNADJUSTED aggregate mRNA coupling â€” the DENOMINATOR of `comp_tcga_mrna_driver_ret`. AUTHORED

comp_tcga_mrna_top_budget
GENE | CATEGORICAL | 97%

The compartment the gene's miRNA BUDGET loads on most â€” a property of the REGULATORS. AUTHORED

comp_tcga_mrna_top_budget_r
GENE | SIGNED Â^'1Â€¦1 | 97%

TCGA Â· the top-ranked Â· pressure budget Â· correlation DERIVED

comp_tcga_mrna_top_target
GENE | CATEGORICAL | 97%

The compartment the TARGET GENE's expression loads on most. AUTHORED

comp_tcga_mrna_top_target_r
GENE | SIGNED Â˘'1Â€¦1 | 97%

TCGA Â· the top-ranked Â· correlation DERIVED

IDENTITY_ Â· 4

reference â€" context, provenance and rarely-quoted detail

4 columns Â· coverage **95%â€"97%** (median 95%)
2× fraction 0â€"1 Â· 1× real, signed Â· 1× boolean

identity_abs
FAMILY | FRACTION 0Â€"1 | 95%

|identity| / Î£|identity| per gene â€" the BOUNDED companion to `identity`, always in [0,1] (verified: 0 non-NaN values outside it on either card). AUTHORED

identity_coherence
FAMILY | FRACTION 0Â€"1 | 95%

1 / Î£|identity| per GENE, in (0,1] â€" identity's own sign coherence. 1.0 when nothing cancels, â†' 0 as suppressor cancellation grows. AUTHORED

identity_deconv
FAMILY | REAL, SIGNED | 97%

Shapley/LMG attribution on R^2 with bagged-NNLS weights â€" WHO regulates the gene, collinearity-fair. AUTHORED

identity_reliable
FAMILY | BOOLEAN | 95%

`identity_coherence â‰¥ 0.5` AND `|identity| â‰¤ 1`. Admits **93.3%** of edge rows and **92.6%** of gene_family rows, and excludes **every** row whose |identity| exceeds 1 (119/119 and 118/118). AUTHORED

PIP_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· coverage **100%â€"100%** (median 100%)
1× constant Â· 1× fraction 0â€"1

pip_dense
FAMILY | CONSTANT | 100%

âš CONSTANT by construction (Ï€â‰¡1 dense readout). AUTHORED

pip_discovery
FAMILY | FRACTION 0Â€"1 | 100%

Posterior inclusion probability. `pip_dense` comes from the pi=1 coupling readout; `pip_discovery` from the evidence-pi readout. AUTHORED

W_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€"100%** (median 100%)
1× real â‰¥ 0

w_max
GENE | REAL Â‰¥ 0 | 100%

Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card. AUTHORED

M_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

m_nnls
FAMILY | REAL â‰¥ 0 | 100%

Bagged-NNLS weight for this (gene, family) cell. AUTHORED

COMPOSITION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× categorical

composition_class
FAMILY | CATEGORICAL | 100%

Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block). AUTHORED

PRIOR_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× fraction 0â€“1

prior_pi
FAMILY | FRACTION 0â€“1 | 100%

The evidence prior fed to the discovery readout. AUTHORED

RETENTION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· coverage **100%â€“100%** (median 100%)
1× boolean

retention_reliable
FAMILY | BOOLEAN | 100%

Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED arm_card.tsv

arm

One row per the miRNA arm itself, gene-free. The deepest card. An arm-rung column must be constant within arm ACROSS genes; if it varies by gene it is not an arm property, whatever its name says.

296 columns **2,450** rows **43** blocks median coverage **30%**

Contents â€" 43 blocks, most important first

ADM_ Â· 4

core â€" the columns findings rest on

4 columns Â· coverage **14%â€"29%** (median 29%)
3× count Â· 1× fraction 0â€"1

adm_frac_with_site
ARM FRACTION 0â€"1 14%

`adm_n_with_site / adm_n_edges`. AUTHORED

adm_n_admissible
ARM COUNT 29%

Edges of this arm that pass admissibility (`adm_has_site` + the evidence bar) â€" MH-216's single most load-bearing conditioning variable, rolled up per arm. AUTHORED

adm_n_edges
ARM COUNT 29%

ADMISSIBILITY â€" could this edge repress in breast AT ALL? `has_site` (seed match in the 3'UTR) AND `expressed` (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED

adm_n_with_site
ARM COUNT 29%

Edges of this arm whose target carries a seed site â€” **89.9% of edges (4,436/4,937)**, so the site requirement is NOT the binding constraint; the evidence bar is (admissible drops it to 60.4%). 19.7% of arms have none. AUTHORED

> *Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% Â· expressed 63.9% Â· admissible 60.4%)*

ARM_ Â· 7

core â€” the columns findings rest on

---

7 columns Â· coverage **15%â€“20%** (median 20%)
4× real, signed Â· 2× real â‰¥ 0 Â· 1× percent

arm_iqr
ARM REAL â‰¥ 0 20%

Interquartile range of the arm's tumour RPM â€” its DYNAMIC RANGE. AUTHORED

---

arm_lfc_HLY_NAT_raw
ARM REAL, SIGNED 15%

log-FC GTEx-healthy â†’ TCGA-NAT, raw. AUTHORED

---

arm_lfc_HLY_TUM_QN
ARM REAL, SIGNED 16%

log-FC GTEx-healthy â†’ TCGA-tumour, on quantile-normalised values. AUTHORED

---

arm_lfc_HLY_TUM_raw
ARM REAL, SIGNED 15%

As `arm_lfc_HLY_TUM_QN` but against the RAW GTEx median. AUTHORED

---

arm_lfc_NAT_TUM
ARM REAL, SIGNED 20%

log-FC of the arm's median abundance, matched NAT â†’ tumour. AUTHORED

---

arm_med_rpm
ARM REAL â‰¥ 0 20%

The arm's MEDIAN RPM across tumour samples â€” its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data. AUTHORED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_pct_above_floor
ARM PERCENT 20%

â›” **THE NAME READS BACKWARDS.** Computed as `100Â·mean(x > FLOOR)` â€” the percent of samples ABOVE the detection floor, i.e. AUTHORED

SEED_ Â· 1

core â€” the columns findings rest on

---

1 columns Â· coverage **100%â€“100%** (median 100%)
1× identifier

seed_family
ARM IDENTIFIER 100%

KEY â€” the seed family itself, one row per family. AUTHORED

SITE_ Â· 12

supporting â€” read these when the core raises a question

Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.
12 columns Â· coverage **29%â€“29%** (median 29%) Â· **7** share the block description
7× count Â· 2× fraction 0â€“1 Â· 2× real, signed Â· 1× real â‰¥ 0

site_frac_8mer
ARM FRACTION 0â€“1 29%

8mer share of the arm's **CANONICAL** sites (`site_n_8mer / site_n_canonical`; median 0.150). AUTHORED

---

site_frac_canonical
ARM | FRACTION 0â€“1 | 29%

Canonical share of the arm's **TOTAL** sites (`site_n_canonical / site_n_total` ; median 0.054) â€” a different denominator from `site_frac_8mer` . AUTHORED

site_n_6mer
ARM | COUNT | AGG EDGE | 29%

count Â· 6mer sites DERIVED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_n_7mer
ARM | COUNT | AGG EDGE | 29%

count Â· 7mer sites DERIVED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_n_8mer
ARM | COUNT | AGG EDGE | 29%

count Â· 8mer sites DERIVED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_n_canonical
ARM | COUNT | AGG EDGE | 29%

count Â· canonical sites DERIVED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_n_genes_canonical
ARM | COUNT | AGG EDGE | 29%

count Â· genes Â· canonical sites DERIVED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_n_noncanon
ARM | COUNT | AGG EDGE | 29%

count Â· non-canonical sites DERIVED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_n_total
ARM | COUNT | AGG EDGE | 29%

count Â· total DERIVED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_repression_med
ARM | REAL, SIGNED | AGG EDGE | 29%

Median predicted repression score over the arm's sites (all negative by construction; median âˆ’0.316). A PREDICTION from sequence, not a measurement â€” never cite it as evidence of observed repression. AUTHORED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

site_repression_min
ARM | REAL, SIGNED | AGG EDGE | 29%

Strongest (most negative) predicted repression among the arm's sites (median âˆ’1.964). AUTHORED

site_sites_per_gene
ARM | REAL â‰¥ 0 | 29%

Canonical sites per targeted gene (median 1.34) â€” site DENSITY, separating an arm with many sites on few genes from one spread thin. `site_n_canonical / site_n_genes_canonical` . AUTHORED

> *Defined on: arms in the scanMiR site-type table (771) â€” âš HALLMARK-SCOPED, 1,432 genes only*

ARB_ Â· 16

supporting â€” read these when the core raises a question

> Per-ARM rollup of its edge set â€” how many edges and genes it regulates and its mean |beta| over them.
>
> **16** columns Â· coverage **28%â€“30%** (median 30%) Â· **10** share the block description
> 12× count Â· 2× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× signed âˆ’1â€¦1

arb_frac_identified
ARM | FRACTION 0â€“1 | AGG EDGE | 30%

`arb_n_identified / arb_n_edges` (verified exact on 100% of rows). AUTHORED

arb_max_abs_beta
ARM | REAL â‰¥ 0 | AGG EDGE | 30%

Largest |Î²| among the arm's edges. AUTHORED

arb_max_identity
ARM · SIGNED âˆ’1â€¦1 · AGG EDGE · 28%

Largest `identity` among this arm's edges, over reliable rows only. âœ… Now runs **âˆ’0.255 â€¦ +1.000** â€” it reached **+740.0** until the inert `beta_frac_reliable` gate was replaced (2026-08-19). AUTHORED

arb_mean_abs_beta
ARM · REAL â‰¥ 0 · AGG EDGE · 30%

Mean |Î²| over the arm's edges. See `arb_max_abs_beta` for the one-edge degeneracy (9.3% of arms, where the two are equal by construction). AUTHORED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b001
ARM · COUNT · 30%

count Â· at the 0.001 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b001_ident
ARM · COUNT · 30%

count Â· at the 0.001 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b005
ARM · COUNT · 30%

count Â· at the 0.005 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b005_ident
ARM · COUNT · 30%

count Â· at the 0.005 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b010
ARM · COUNT · 30%

count Â· at the 0.010 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b010_ident
ARM · COUNT · 30%

count Â· at the 0.010 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b025
ARM · COUNT · 30%

count Â· at the 0.025 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b025_ident
ARM · COUNT · 30%

count Â· at the 0.025 threshold DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_comp_explained
ARM · COUNT · AGG EDGE · 30%

count DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_edges
ARM · COUNT · AGG EDGE · 30%

Edges this arm carries in the LEARNED MODEL's own readouts. AUTHORED

arb_n_identified
ARM · COUNT · AGG EDGE · 30%

count DERIVED

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_identity_reliable
ARM · COUNT · AGG EDGE · 30%

Edges of this arm whose `identity` share is usable (`identity_reliable`). AUTHORED

HEALTHY_ Â· 3

supporting â€” read these when the core raises a question

> Healthy-state legs and flags â€” including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim.
>
> **3** columns Â· coverage **16%â€“20%** (median 16%) Â· **3** share the block description
> 1× categorical Â· 1× real â‰¥ 0 Â· 1× boolean

healthy_leg
ARM | CATEGORICAL | 20%

Healthy-state legs and flags â€" including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim. BLOCK-LEVEL

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

healthy_potential
ARM | REAL Â‰¥ 0 | 16%

Healthy-state legs and flags â€" including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim. BLOCK-LEVEL

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

healthy_uninformative
ARM | BOOLEAN | 16%

Healthy-state legs and flags â€" including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim. BLOCK-LEVEL

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

(BARE) Â· 4

supporting â€" read these when the core raises a question

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.

4 columns Â· coverage **15%â€"100%** (median 17%) Â· **3** share the block description
3× real, signed Â· 1× identifier

arm
KEY | IDENTIFIER | 100%

KEY â€" mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED

dGlobal_HLY_NAT
ARM | REAL, SIGNED | 15%

in the GTEx-healthy state Â· in matched normal (NAT) DERIVED

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

dGlobal_HLY_TUM
ARM | REAL, SIGNED | 15%

in the GTEx-healthy state Â· in tumour DERIVED

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

dGlobal_NAT_TUM
ARM | REAL, SIGNED | 20%

in matched normal (NAT) Â· in tumour DERIVED

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

FAME_ Â· 28

supporting â€" read these when the core raises a question

Study-attention axis for the arm â€" distinct PMIDs and distinct curated genes.

28 columns Â· coverage **94%â€"96%** (median 96%) Â· **23** share the block description
27× count Â· 1× fraction 0â€"1

fame_assay_binding__functional_mti_weak_studies
ARM | COUNT | 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· weak evidence Â· study count DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

fame_assay_binding_studies
ARM | COUNT | 96%

Studies whose assay class is BINDING (the marginal of the assay Ã— functional-class cross-tab). AUTHORED

fame_assay_functional_mti_studies
ARM | COUNT | 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· study count DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

fame_assay_functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€“target interaction evidence Â· weak evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_nonfunctional_mti_studies
ARM COUNT 96%

by assay class Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_other__functional_mti_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_other__functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€“target interaction evidence Â· weak evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_other__nonfunctional_mti_studies
ARM COUNT 96%

by assay class Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_other_studies
ARM COUNT 96%

by assay class Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_protein__functional_mti_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_protein__functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€“target interaction evidence Â· weak evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_protein__nonfunctional_mti_studies
ARM COUNT 96%

by assay class Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_protein_studies
ARM COUNT 96%

by assay class Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_reporter__functional_mti_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_reporter__nonfunctional_mti_studies
ARM COUNT 96%

by assay class Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_reporter_studies
ARM COUNT 96%

by assay class Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_rna__functional_mti_studies
ARM COUNT 96%

by assay class Â· against mRNA Â· functional assays Â· miRNAâ€“target interaction evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_rna__functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· against mRNA Â· functional assays Â· miRNAâ€“target interaction evidence Â· weak evidence Â· study count DERIVED

Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)

fame_assay_rna__nonfunctional_mti_studies

ARM COUNT 96%

by assay class Â· against mRNA Â· miRNAâ€“target interaction evidence Â· study count DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_rna_studies

ARM COUNT 96%

by assay class Â· against mRNA Â· study count DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_total_studies

ARM COUNT 96%

Distinct STUDIES (`study_id.nunique()`) behind this arm's curated edges â€" median 57, max 930. See `fame_assay_unique_targets` : different definitions, one axis (spearman 1.0000). AUTHORED

---

fame_assay_unique_targets

ARM COUNT 96%

Distinct TARGET GENES this arm has in the curated literature (`gene.nunique()`). AUTHORED

---

fame_led_frac_ltfunc

ARM FRACTION 0â€“1 AGG EDGE 96%

fraction DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_led_n_classes

ARM COUNT AGG EDGE 96%

How many distinct evidence CLASSES this arm's curated edges span (1â€“7). AUTHORED

---

fame_led_n_ltfunc_pmid

ARM COUNT AGG EDGE 96%

count Â· distinct publications DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_led_n_weak

ARM COUNT AGG EDGE 94%

count Â· weak evidence DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_n_genes_curated

ARM COUNT AGG EDGE 96%

count Â· genes DERIVED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_npmid

ARM COUNT AGG EDGE 96%

Distinct PMIDs citing this arm. AUTHORED

AGO_ Â· 4

supporting â€" read these when the core raises a question

> **4** columns Â· coverage **28%â€“60%** (median 28%)
> 1× fraction 0â€“1 Â· 1× constant Â· 1× categorical Â· 1× real â‰¥ 0

ago_dom

ARM FRACTION 0â€“1 28%

Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm â€" the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage. AUTHORED

---

ago_dom_src

ARM CONSTANT 28%

Provenance of `ago_dom` â€" which CLIP dataset the AGO-loading dominance came from. AUTHORED

---

ago_guide_class

ARM CATEGORICAL 28%

AGO/RISC loading gate for the arm. AUTHORED

---

ago_reads

ARM REAL â‰¥ 0 60%

Raw AGO-CLIP read count (median 6, max 210,722). AUTHORED

HLY_ Â· 7

supporting â€” read these when the core raises a question

HEALTHY baseline levels for the arm â€” GTEx breast median/percentile and TCGA-NAT median.

7 columns Â· coverage **17%â€“91%** (median 91%) Â· **5** share the block description

2× real â‰¥ 0 Â· 1× categorical Â· 1× boolean Â· 1× percent Â· 1× count Â· 1× fraction 0â€“1

hly_baseline_src
ARM CATEGORICAL 91%

source DERIVED

Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value

hly_from_seedmate
ARM BOOLEAN 91%

HEALTHY baseline levels for the arm â€” GTEx breast median/percentile and TCGA-NAT median. BLOCK-LEVEL

Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value

hly_gtex_median
ARM REAL â‰¥ 0 17%

GTEx Â· median DERIVED

Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value

hly_gtex_pct
ARM PERCENT 17%

GTEx Â· percent DERIVED

Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value

hly_nat_detect_n
ARM COUNT 91%

In how many of the ~104 matched NAT samples this arm is detected above the floor. AUTHORED

hly_nat_median
ARM REAL â‰¥ 0 74%

Median NAT abundance (fill 74.0%). AUTHORED

hly_tumor_prev
ARM FRACTION 0â€“1 91%

HEALTHY baseline levels for the arm â€” GTEx breast median/percentile and TCGA-NAT median. BLOCK-LEVEL

Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value

DOSE_ Â· 3

supporting â€” read these when the core raises a question

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.

3 columns Â· coverage **12%â€“12%** (median 12%) Â· **3** share the block description

2× real â‰¥ 0 Â· 1× constant

dose_comp_retention
ARM REAL â‰¥ 0 12%

retention DERIVED

Defined on: edges with a matched NAT leg (n~104 paired participants)

dose_confounded
ARM CONSTANT 12%

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. BLOCK-LEVEL

Defined on: edges with a matched NAT leg (n~104 paired participants)

dose_prolif_retention
ARM REAL â‰¥ 0 12%

retention DERIVED

Defined on: edges with a matched NAT leg (n~104 paired participants)

ABUND_ Â· 5

supporting â€” read these when the core raises a question

The arm's own abundance profile across the cohort â€” median, SD, IQR, detection fraction.

5 columns Â· coverage **88%â€“91%** (median 91%) Â· **4** share the block description

3× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× categorical

abund_frac_detected
ARM | FRACTION 0â€“1 | 91%

fraction DERIVED

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

abund_iqr
ARM | REAL â‰¥ 0 | 91%

interquartile range DERIVED

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

abund_med
ARM | REAL â‰¥ 0 | 91%

median DERIVED

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

abund_sd
ARM | REAL â‰¥ 0 | 88%

standard deviation DERIVED

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

abund_sd_tertile
ARM | CATEGORICAL | 88%

Dispersion tertile â€” **718 / 716 / 718 by construction**, so it carries no information beyond `abund_sd`'s rank. AUTHORED

ATTR_ Â· 14

reference â€” context, provenance and rarely-quoted detail

Attribution rollup over the arm's edges â€” how much of the gene's coupling this arm is credited with, and over how many reliable edges.

14 columns Â· coverage **6%â€“29%** (median 15%) Â· **9** share the block description

11× fraction 0â€“1 Â· 2× count Â· 1× signed âˆ’1â€¦1

attr_max_budget_share
ARM | FRACTION 0â€“1 | 15%

maximum Â· pressure budget Â· share DERIVED

Defined on: arms with a Shapley/attribution decomposition over their edges (728)

attr_max_credit_share
ARM | FRACTION 0â€“1 | 6%

maximum Â· share DERIVED

Defined on: arms with a Shapley/attribution decomposition over their edges (728)

attr_max_joint_share
ARM | FRACTION 0â€“1 | 15%

Largest JOINT attribution share among this arm's edges. AUTHORED

attr_max_nested_share
ARM | FRACTION 0â€“1 | 15%

Largest NESTED attribution share. See `attr_max_joint_share`: different values (agree on 35.4%), one axis (spearman +0.962). AUTHORED

Defined on: arms with a Shapley/attribution decomposition over their edges (728)

attr_max_within_share
ARM | FRACTION 0â€“1 | 15%

Largest WITHIN-cell attribution share for this arm. AUTHORED

attr_med_budget_share
ARM | FRACTION 0â€“1 | 15%

median Â· pressure budget Â· share DERIVED

Defined on: arms with a Shapley/attribution decomposition over their edges (728)

attr_med_credit_share
ARM | FRACTION 0â€“1 | 6%

median Â· share DERIVED

Defined on: arms with a Shapley/attribution decomposition over their edges (728)

attr_med_dose_share
ARM | FRACTION 0â€“1 | 29%

Median dose share of this arm within its family cells. AUTHORED

attr_med_joint_share
ARM | FRACTION 0â€“1 | 15%

median Â· share DERIVED

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

attr_med_nested_share
ARM | FRACTION 0â€“1 | 15%

median Â· share DERIVED

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

attr_med_phi_joint
ARM | SIGNED Â˜’1â€¦1 | 15%

median DERIVED

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

attr_med_within_share
ARM | FRACTION 0â€“1 | 15%

median Â· share DERIVED

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

attr_n_edges
ARM | COUNT | 22%

count DERIVED

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

attr_n_reliable
ARM | COUNT | 15%

Reliable edges behind this arm's `attr_*` shares (median 2). AUTHORED

CTX_ Â· 2

reference â€" context, provenance and rarely-quoted detail

> Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
>
> **2** columns Â· coverage **29%â€“29%** (median 29%) Â· **2** share the block description
> 1× boolean Â· 1× real â‰¥ 0

ctx_arm_abundant
ARM | BOOLEAN | 29%

per arm DERIVED

> *Defined on: arms present on the EDGE card (729 of 2,450) â€" lifted arm-rung columns, names preserved*

ctx_arm_dose
ARM | REAL â‰¥ 0 | 29%

per arm Â· dose DERIVED

> *Defined on: arms present on the EDGE card (729 of 2,450) â€" lifted arm-rung columns, names preserved*

GRANK_ Â· 3

reference â€" context, provenance and rarely-quoted detail

> The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.
>
> **3** columns Â· coverage **15%â€“20%** (median 20%) Â· **3** share the block description
> 3× count

grank_HLY
ARM | COUNT | 15%

in the GTEx-healthy state DERIVED

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

grank_NAT
ARM | COUNT | 20%

in matched normal (NAT) DERIVED

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

grank_TUM
ARM | COUNT | 20%

in tumour DERIVED

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

REAL_ Â· 11

reference â€” context, provenance and rarely-quoted detail

> Realization rollup for the ARM â€” how many of its edges are scored, and its median and best realized coupling.
>
> **11** columns Â· coverage **9%â€“24%** (median 24%) Â· **8** share the block description
>
> 7× count Â· 2× signed â'1â€¦1 Â· 2× fraction 0â€“1

real_best_coupling
ARM · SIGNED Âˆ'1Â€¦1 · 24%

Most negative realized coupling among this arm's scored genes (median âˆ'0.08, min âˆ'0.56). AUTHORED

real_frac_c10
ARM · FRACTION 0Â€“1 · 9%

Fraction of this arm's scored genes coupling below âˆ'0.10. AUTHORED

real_frac_sig
ARM · FRACTION 0Â€“1 · 9%

fraction DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_med_coupling
ARM · SIGNED Âˆ'1Â€¦1 · 24%

median DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_n_c05
ARM · COUNT · 24%

count Â· at the âˆ'0.05 threshold DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_n_c10
ARM · COUNT · 24%

count Â· at the âˆ'0.10 threshold DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_n_c15
ARM · COUNT · 24%

count Â· at the âˆ'0.15 threshold DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_n_c20
ARM · COUNT · 24%

count Â· at the âˆ'0.20 threshold DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_n_c30
ARM · COUNT · 24%

count Â· at the âˆ'0.30 threshold DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_n_scored
ARM · COUNT · 24%

Genes scored for this arm's realized coupling (median 3). âœ… Its `n_c05 â€¦ n_c30` ladder is properly nested â€" **0 violations** (verified). AUTHORED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

real_n_sig
ARM · COUNT · 24%

count DERIVED

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

TS_ Â· 10

reference â€" context, provenance and rarely-quoted detail

> TargetScan context++ scores for the arm's sites â€" number of sites, mean and best context score. More negative context++ = stronger predicted repression.
>
> **10** columns Â· coverage **13%â€“13%** (median 13%) Â· **8** share the block description
>
> 5× count Â· 2× real, signed Â· 1× signed âˆ'1â€¦1 Â· 1× fraction 0â€“1 Â· 1× real â‰¥ 0

ts_ctx_best
ARM | REAL, SIGNED | 13%

Best (most negative) TargetScan context++ score for this arm â€” range [âˆ’2.02, âˆ’0.34], **all negative by construction** (context++ scores repression as a negative number). AUTHORED

ts_ctx_mean
ARM | SIGNED âˆ’1â€¦1 | 13%

mean DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ts_frac_8mer
ARM | FRACTION 0â€“1 | 13%

fraction Â· 8mer sites DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ts_kd_med
ARM | REAL, SIGNED | 13%

median DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ts_n_7mer_A1
ARM | COUNT | 13%

count Â· 7mer sites DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ts_n_7mer_m8
ARM | COUNT | 13%

count Â· 7mer sites DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ts_n_8mer
ARM | COUNT | 13%

count Â· 8mer sites DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ts_n_genes
ARM | COUNT | AGG EDGE | 13%

Genes with a TargetScan context++ site â€” median 497, fill **12.9%, the SPARSEST universe on the card**. Correlates +0.726 with the MANE seed scan and only **+0.231** with scanMiR-any. AUTHORED

ts_n_sites
ARM | COUNT | AGG EDGE | 13%

count DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ts_sites_per_gene
ARM | REAL â‰¥ 0 | 13%

per gene DERIVED

Defined on: arms in TargetScan's human default predictions (âš only 321)

ISOC_ Â· 5

reference â€” context, provenance and rarely-quoted detail

> isomiR COMPOSITION â€” what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm.
> **5** columns Â· coverage **91%â€“91%** (median 91%) Â· **4** share the block description
> 3× fraction 0â€“1 Â· 1× real â‰¥ 0 Â· 1× count

isoc_canonical_frac
ARM | FRACTION 0â€“1 | 91%

canonical sites Â· fraction DERIVED

Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss

isoc_donated_frac
ARM | FRACTION 0â€“1 | 91%

fraction DERIVED

Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss

isoc_med_reads
ARM | REAL â‰¥ 0 | 91%

Median isomiR reads for this arm â€” **median 1.00, max 528,927**. AUTHORED

isoc_n_dest_families
ARM | COUNT | 91%

count DERIVED

Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss

isoc_orphan_frac
ARM | FRACTION 0â€“1 | 91%

fraction DERIVED

Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss

SDSZ_ Â· 4

reference â€” context, provenance and rarely-quoted detail

> MANE 3'UTR seed scan.
>
> **4** columns Â· coverage **97%â€“97%** (median 97%) Â· **3** share the block description
> 2× count Â· 1× real â‰¥ 0 Â· 1× fraction 0â€“1

sdsz_N_7mer_plus
ARM | COUNT | AGG EDGE | 97%

count Â· 7mer sites DERIVED

> *Defined on: arms in the MANE 3'UTR seed scan â€” âš a FOURTH targetome universe, never blend (2,370)*

sdsz_N_8mer
ARM | COUNT | AGG EDGE | 97%

count Â· 8mer sites DERIVED

> *Defined on: arms in the MANE 3'UTR seed scan â€" âš a FOURTH targetome universe, never blend (2,370)*

sdsz_N_eff
ARM | REAL â‰¥ 0 | AGG EDGE | 97%

Effective seed-site count over the scanned universe (median 2,094, fill **96.7%** â€” the best-covered targetome axis). AUTHORED

sdsz_rarity
ARM | FRACTION 0â€“1 | 97%

MANE 3'UTR seed scan. BLOCK-LEVEL

> *Defined on: arms in the MANE 3'UTR seed scan â€" âš a FOURTH targetome universe, never blend (2,370)*

CHIM_ Â· 5

reference â€” context, provenance and rarely-quoted detail

> Chimeric-evidence ROLLUP for the arm (aggregated over its edges).
>
> **5** columns Â· coverage **21%â€“63%** (median 60%) Â· **5** share the block description
> 4× count Â· 1× real â‰¥ 0

chim_manakov_n_genes
ARM | COUNT | AGG EDGE | 60%

count Â· genes DERIVED

> *Defined on: arms with chimeric ligation evidence (1,538) â€” âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_manakov_weight
ARM | REAL â‰¥ 0 | AGG EDGE | 60%

Chimeric-evidence ROLLUP for the arm (aggregated over its edges). BLOCK-LEVEL

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_n_sources
ARM | COUNT | AGG EDGE | 63%

count DERIVED

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_tarbase_n_genes
ARM | COUNT | AGG EDGE | 21%

count Â· genes DERIVED

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_tarbase_weight
ARM | COUNT | AGG EDGE | 21%

Chimeric-evidence ROLLUP for the arm (aggregated over its edges). BLOCK-LEVEL

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

FST_ Â· 16

reference â€” context, provenance and rarely-quoted detail

â THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES â€” gene-free (`card_ladders.family_state_shift`, 2026-08-19).

16 columns Â· coverage **3%â€“100%** (median 20%) Â· **8** share the block description
6× count Â· 5× boolean Â· 3× fraction 0â€“1 Â· 2× signed âˆ’1â€¦1

fst_d_share_HLY_TUM ARM · SIGNED âˆ’1â€¦1 · 3%	As `fst_d_share_NAT_TUM` but GTEx-healthy â†' tumour, same common-support correction. n=68 (HLY is the sparsest leg at 17.0%). Post-fix **mean âˆ'0.0000, 47.1% positive, p=0.93**. AUTHORED
fst_d_share_NAT_TUM ARM · SIGNED âˆ’1â€¦1 · 6%	Change in the arm's share of its seed family's linear dose, matched NAT â†' tumour, **re-normalised over the members measured in BOTH states**. AUTHORED
fst_dominance_switch ARM · BOOLEAN · 7%	â THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES â€” gene-free (`card_ladders.family_state_shift`, 2026-08-19). BLOCK-LEVEL
fst_hly_family_measured ARM · BOOLEAN · 100%	Do ALL members of this arm's family have a measured GTEx level. AUTHORED
fst_hly_measured ARM · BOOLEAN · 100%	in the GTEx-healthy state DERIVED
fst_is_dominant_HLY ARM · BOOLEAN · 17%	the dominant one Â· in the GTEx-healthy state DERIVED
fst_is_dominant_TUM ARM · BOOLEAN · 20%	the dominant one Â· in tumour DERIVED
fst_n_common_HLY_TUM ARM · COUNT · 9%	Members measured in BOTH GTEx-healthy and tumour â€” the denominator behind `fst_d_share_HLY_TUM`. See `fst_n_common_NAT_TUM`. AUTHORED
fst_n_common_NAT_TUM ARM · COUNT · 20%	How many of the family's arms are measured in BOTH NAT and tumour â€” the DENOMINATOR behind `fst_d_share_NAT_TUM`, shipped because the column was wrong for want of it. AUTHORED
fst_n_members ARM · COUNT · 100%	count Â· family members DERIVED
fst_rank_HLY ARM · COUNT · 17%	rank Â· in the GTEx-healthy state DERIVED
fst_rank_NAT ARM · COUNT · 74%	rank Â· in matched normal (NAT) DERIVED
fst_rank_TUM ARM · COUNT · 20%	rank Â· in tumour DERIVED
fst_share_HLY ARM · FRACTION 0â€“1 · 17%	Share of family dose in GTEx-healthy, from `hly_gtex_median` (fill **17.0%** â€” the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only. AUTHORED
fst_share_NAT ARM · FRACTION 0â€“1 · 74%	Share of family dose in matched NAT, from `hly_nat_median` (fill **74.0%** â€” the best-covered leg, 3.7Ã— TUM). Same singleton degeneracy: median 1.000, mask on `fst_n_members > 1`. AUTHORED

fst_share_TUM
ARM | FRACTION 0â€“1 | 20%

The arm's fraction of its seed family's total linear dose in tumour, from `arm_med_rpm` (fill **19.8%**). AUTHORED

SUB_ Â· 6

reference â€” context, provenance and rarely-quoted detail

> PAM50 subtype block aggregated to the ARM. See `esub_` for the per-edge form.
> **6** columns Â· coverage **30%â€“30%** (median 30%) Â· **4** share the block description
> 2× count Â· 2× signed âˆ’1â€¦1 Â· 1× categorical Â· 1× fraction 0â€“1

sub_best_frac
ARM | FRACTION 0â€“1 | 30%

the best Â· fraction DERIVED

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_best_mode
ARM | CATEGORICAL | 30%

the best DERIVED

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_med_rho_best
ARM | SIGNED âˆ’1â€¦1 | 30%

Median coupling in the arm's BEST subtype. AUTHORED

sub_med_rho_pool
ARM | SIGNED âˆ’1â€¦1 | 30%

Median pooled-subtype coupling (median âˆ’0.0064) â€" the unselected reference for `sub_med_rho_best`. AUTHORED

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_n_edges
ARM | COUNT | 30%

count DERIVED

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_n_q05
ARM | COUNT | 30%

count DERIVED

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

SEQ_ Â· 4

reference â€” context, provenance and rarely-quoted detail

> SEQUENCE targetome degree â€" genes with a strong predicted duplex (scanMiR RBNS K_D), breast-expressed.
> **4** columns Â· coverage **30%â€“30%** (median 30%) Â· **2** share the block description
> 2× count Â· 2× real â‰¥ 0

seq_n_genes_any
ARM | COUNT | AGG EDGE | 30%

Genes with ANY scanMiR K_D site â€" AUTHORED

seq_n_genes_strong
ARM | COUNT | AGG EDGE | 30%

Genes with a STRONG scanMiR K_D site â€" median 3,632, spearman **+0.822** with `sdsz_N_eff` and **+0.797** with `site_n_genes_canonical`, but only **+0.566** with `ts_n_genes`. AUTHORED

seq_promisc
ARM | REAL â‰¥ 0 | 30%

SEQUENCE targetome degree â€" genes with a strong predicted duplex (scanMiR RBNS K_D), breast-expressed. BLOCK-LEVEL

> *Defined on: arms in the scanMiR GENOME-WIDE K_D scan (746; breast-expressed genes)*

seq_promisc_any
ARM | REAL â‰¥ 0 | 30%

SEQUENCE targetome degree â€" genes with a strong predicted duplex (scanMiR RBNS K_D), breast-expressed. BLOCK-LEVEL

> *Defined on: arms in the scanMiR GENOME-WIDE K_D scan (746; breast-expressed genes)*

CUR_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· coverage **36%â€“36%** (median 36%)
2× count

cur_he_degree
ARM COUNT AGG EDGE 36%

How many HE genes this arm regulates in the curated design â€” the arm's curation degree (median 3, max 125). AUTHORED

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)*

cur_he_degree_expr
ARM COUNT AGG EDGE 36%

As `cur_he_degree` but counting only targets above the expression floor. AUTHORED

COV_ Â· 22

reference â€” context, provenance and rarely-quoted detail

Coverage flags â€” was this row ever SCANNED by the named source.

22 columns Â· coverage **100%â€“100%** (median 100%) Â· **21** share the block description
22× boolean

cov_ago_dom
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_attr
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_beta
ARM BOOLEAN 100%

Î² DERIVED

cov_cnv
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_comp
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_cptac
ARM BOOLEAN 100%

CPTAC DERIVED

> *Defined on: arms measurable in CPTAC (2,095)*

cov_curated
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_expr
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_famrole
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_field
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_focal
ARM BOOLEAN 100%

Coverage flags â€” was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_fst
ARM BOOLEAN 100%

Was the family-share block computed for this arm at all â€” i.e. did it have the abundance input. AUTHORED

cov_gctx
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_isocomp
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_isomir
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_meth
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_real
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_seed_scan
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_seq_scan
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_site_types
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_subtype
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

cov_targetscan
ARM BOOLEAN 100%
Coverage flags â€" was this row ever SCANNED by the named source. BLOCK-LEVEL

ISO_ Â· 5

reference â€" context, provenance and rarely-quoted detail

> isomiR profile for the arm â€" 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional).
> **5** columns Â· coverage **28%â€"28%** (median 28%) Â· **4** share the block description
> 2× fraction 0â€"1 Â· 1× count Â· 1× real â‰¥ 0 Â· 1× real, signed

iso_n_samples
ARM COUNT 28%
count DERIVED
> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

iso_rpm_summed
ARM REAL â‰¥ 0 28%
âš **A SUM ACROSS SAMPLES, not a per-sample RPM â€" the name misleads.** Max **270,988,199**, which is impossible for a per-million unit; divided by `iso_n_samples` the scale is sane (median **57**, max 251,381). AUTHORED

iso_seed_shift_frac
ARM FRACTION 0â€"1 28%
shift Â· fraction DERIVED
> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

iso_seed_shift_sd
ARM FRACTION 0â€"1 28%
shift Â· standard deviation DERIVED
> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

iso_top_shift_nt
ARM REAL, SIGNED 28%
the top-ranked Â· shift DERIVED
> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

FIELD_ Â· 5

reference â€” context, provenance and rarely-quoted detail

> Within-patient NAT field-effect fit â€” is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset.
>
> **5** columns Â· coverage **23%â€“23%** (median 23%) Â· **3** share the block description
>
> 4× signed âˆ’1â€¦1 Â· 1× count

field_excess_over_perm
ARM | SIGNED âˆ’1â€¦1 | 23%

â›”â›” **NOT A RETENTION RATIO â€” a FIFTH sense of the word in this codebase.** Measured: it equals **`field_r_own âˆ’ field_r_perm`** (exact within rounding on 100% of 571 arms, max residual 0.001), i.e. AUTHORED

field_n
ARM | COUNT | 23%

count DERIVED

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

field_r_adj
ARM | SIGNED âˆ’1â€¦1 | 23%

correlation Â· composition-adjusted DERIVED

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

field_r_own
ARM | SIGNED âˆ’1â€¦1 | 23%

correlation DERIVED

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

field_r_perm
ARM | SIGNED âˆ’1â€¦1 | 23%

The permutation null for `field_r_own` â€” median **0.000**, range [âˆ’0.11, +0.09]. AUTHORED

FAMROLE_ Â· 6

reference â€” context, provenance and rarely-quoted detail

> The arm's ROLE inside its seed family â€” member count, its share of family abundance, and its rank among members.
>
> **6** columns Â· coverage **88%â€“100%** (median 91%) Â· **5** share the block description
>
> 3× count Â· 1× fraction 0â€“1 Â· 1× categorical Â· 1× boolean

famrole_abund_rank
ARM | COUNT | 91%

unweighted abundance-sum reference Â· rank DERIVED

> *Defined on: arms with a seed family â€” the arm's ROLE inside it; â›” DEGENERATE at famrole_n_members==1*

famrole_abund_share
ARM | FRACTION 0â€“1 | 91%

unweighted abundance-sum reference Â· share DERIVED

> *Defined on: arms with a seed family â€” the arm's ROLE inside it; â›” DEGENERATE at famrole_n_members==1*

famrole_class
ARM | CATEGORICAL | 100%

The arm's role in its seed family â€” **`sole` on 1,639 of 2,450 (66.9%)**, `dominant` 282, `co` and the rest below. AUTHORED

famrole_is_dominant
ARM | BOOLEAN | 91%

the dominant one DERIVED

> *Defined on: arms with a seed family â€” the arm's ROLE inside it; â›” DEGENERATE at famrole_n_members==1*

famrole_n_members
ARM | COUNT | 100%

count Â· family members DERIVED

> *Defined on: arms with a seed family â€” the arm's ROLE inside it; â›” DEGENERATE at famrole_n_members==1*

famrole_var_rank
ARM | COUNT | 88%

rank DERIVED

> *Defined on: arms with a seed family â€” the arm's ROLE inside it; â›” DEGENERATE at famrole_n_members==1*

SURV_ Â· 8

reference â€” context, provenance and rarely-quoted detail

TCGA outcome panel legs.

8 columns Â· coverage **6%â€“6%** (median 6%) Â· **6** share the block description

4× p-value Â· 3× real â‰¥ 0 Â· 1× constant

surv_dfi_hr_adj
ARM | REAL â‰¥ 0 | 6%

Adjusted DFI hazard ratio per arm â€” range **[0.84, 1.16]**, median **1.00**. The whole distribution sits within Â±16% of no effect. See `surv_dfi_q_adj`. AUTHORED

Defined on: arms in the TCGA outcome panel (145) â€” â›” TCGA legs only; the GSE leg is bare-stem matched

surv_dfi_hr_base
ARM | REAL â‰¥ 0 | 6%

TCGA outcome panel legs. BLOCK-LEVEL

Defined on: arms in the TCGA outcome panel (145) â€” â›” TCGA legs only; the GSE leg is bare-stem matched

surv_dfi_p_adj
ARM | P-VALUE | 6%

p-value Â· composition-adjusted DERIVED

Defined on: arms in the TCGA outcome panel (145) â€” â›” TCGA legs only; the GSE leg is bare-stem matched

surv_dfi_p_base
ARM | P-VALUE | 6%

p-value DERIVED

Defined on: arms in the TCGA outcome panel (145) â€” â›” TCGA legs only; the GSE leg is bare-stem matched

surv_dfi_q_adj
ARM | CONSTANT | 6%

BH q for the arm's DFI hazard ratio. AUTHORED

surv_os_hr_adj
ARM | REAL â‰¥ 0 | 6%

composition-adjusted DERIVED

Defined on: arms in the TCGA outcome panel (145) â€” â›” TCGA legs only; the GSE leg is bare-stem matched

surv_os_p_adj
ARM | P-VALUE | 6%

p-value Â· composition-adjusted DERIVED

Defined on: arms in the TCGA outcome panel (145) â€” â›” TCGA legs only; the GSE leg is bare-stem matched

surv_os_q_adj
ARM | P-VALUE | 6%

BH-adjusted q Â· composition-adjusted DERIVED

Defined on: arms in the TCGA outcome panel (145) â€” â›” TCGA legs only; the GSE leg is bare-stem matched

AID_ Â· 4

reference â€” context, provenance and rarely-quoted detail

Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates (`arm_resolvable`, `arm_sep_z`, `oof_drho` medians).

4 columns Â· coverage **6%â€“6%** (median 6%) Â· **3** share the block description

1× fraction 0â€“1 Â· 1× real â‰¥ 0 Â· 1× signed âˆ’1â€¦1 Â· 1× count

aid_frac_resolvable
ARM | FRACTION 0â€“1
AGG ARM-IN-FAMILY | 6%

fraction DERIVED

Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€” the arm rung's 20.3%-of-edges footprint (142)

aid_med_arm_sep_z
ARM | REAL â‰¥ 0
AGG ARM-IN-FAMILY | 6%

median Â· per arm Â· z-score DERIVED

Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€” the arm rung's 20.3%-of-edges footprint (142)

aid_med_oof_drho
ARM | SIGNED Âˆ'1Â€¦1
AGG ARM-IN-FAMILY | 6%

median Â· out-of-fold DERIVED

> *Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€" the arm rung's 20.3%-of-edges footprint (142)*

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aid_n_cells
ARM | COUNT
AGG ARM-IN-FAMILY | 6%

Multi-arm cells this arm participates in â€" fill **5.8%** (the arm identifiability block is the sparsest on the card). AUTHORED

BC_ Â· 11

reference â€" context, provenance and rarely-quoted detail

Breast-cancer locus annotation for the arm â€" CpG probes, CGI overlap and related epigenetic context at the precursor locus.

11 columns Â· coverage **14%â€"36%** (median 14%) Â· **10** share the block description

4× fraction 0â€"1 Â· 4× count Â· 1× boolean Â· 1× signed âˆ'1â€¦1 Â· 1× categorical

bc_fam_context_homogeneous
ARM | BOOLEAN | 36%

per seed family DERIVED

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_dose_coding_host_frac
ARM | FRACTION 0â€"1 | 36%

per seed family Â· dose Â· fraction DERIVED

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_dose_cotranscribed_frac
ARM | FRACTION 0â€"1 | 36%

per seed family Â· dose Â· fraction DERIVED

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_n_distinct_coding_hosts
ARM | COUNT | 36%

per seed family Â· count DERIVED

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_n_members
ARM | COUNT | 36%

per seed family Â· count Â· family members DERIVED

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_meth_dbeta
ARM | SIGNED âˆ'1â€¦1 | 14%

Breast-cancer locus annotation for the arm â€" CpG probes, CGI overlap and related epigenetic context at the precursor locus. BLOCK-LEVEL

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€" 5p and 3p share one promoter and one Î"Î² (354 arms)*

---

bc_meth_direction
ARM | CATEGORICAL | 14%

Tumour-vs-normal methylation direction at this arm's locus. AUTHORED

---

bc_meth_n_cgi
ARM | COUNT | 14%

count DERIVED

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€" 5p and 3p share one promoter and one Î"Î² (354 arms)*

---

bc_meth_n_probes
ARM | COUNT | 14%

count DERIVED

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€" 5p and 3p share one promoter and one Î"Î² (354 arms)*

---

bc_meth_normal_beta
ARM | FRACTION 0â€"1 | 14%

Î² DERIVED

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€" 5p and 3p share one promoter and one Î"Î² (354 arms)*

---

bc_meth_tumour_beta
ARM | FRACTION 0â€"1 | 14%

Î² DERIVED

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€" 5p and 3p share one promoter and one Î"Î² (354 arms)*

WSHIFT_ Â· 6

reference â€" context, provenance and rarely-quoted detail

Within-patient paired quantities on the ~103 matched tumour/NAT participants.

6 columns Â· coverage **45%â€"91%** (median 52%) Â· **6** share the block description

3× real â‰¥ 0 Â· 2× real, signed Â· 1× count

wshift_mean_dGlobalRank
ARM | REAL, SIGNED | 52%

mean DERIVED

Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)

wshift_mean_own_shift
ARM | REAL, SIGNED | 52%

mean Â· shift DERIVED

Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)

wshift_n_pairs
ARM | COUNT | 91%

count DERIVED

Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)

wshift_nat_patient_sd
ARM | REAL â‰¥ 0 | 74%

in matched normal (NAT) Â· standard deviation DERIVED

Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)

wshift_own_specific_frac
ARM | REAL â‰¥ 0 | 45%

fraction DERIVED

Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)

wshift_sd_dGlobalRank
ARM | REAL â‰¥ 0 | 45%

standard deviation DERIVED

Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)

TIER_ Â· 5

reference â€" context, provenance and rarely-quoted detail

Expression tier of the arm against the detection floor â€" class, fraction of samples expressed, p90 and max log2 RPM.

5 columns Â· coverage **91%â€"91%** (median 91%) Â· **4** share the block description

2× real â‰¥ 0 Â· 1× categorical Â· 1× boolean Â· 1× fraction 0â€"1

tier_class
ARM | CATEGORICAL | 91%

Expression tier â€" `silent` **on 1,518 of 2,236 arms (67.9%)**, `conditional` 476, the rest robust. AUTHORED

tier_expressed
ARM | BOOLEAN | 91%

Expression tier of the arm against the detection floor â€" class, fraction of samples expressed, p90 and max log2 RPM. BLOCK-LEVEL

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

tier_frac_expressed
ARM | FRACTION 0â€"1 | 91%

fraction DERIVED

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

tier_max
ARM | REAL â‰¥ 0 | 91%

maximum DERIVED

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

tier_p90
ARM | REAL â‰¥ 0 | 91%

Expression tier of the arm against the detection floor â€" class, fraction of samples expressed, p90 and max log2 RPM. BLOCK-LEVEL

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

DISC_ Â· 2

reference â€” context, provenance and rarely-quoted detail

> **2** columns Â· coverage **1%â€“100%** (median 50%)
> 1× categorical Â· 1× count

disc_gold_families
ARM · CATEGORICAL · 1%
Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

disc_n_gold_edges
ARM · COUNT · 100%
How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p **39** Â· miR-20a-5p 25 Â· miR-93-5p 25 Â· miR-19a-3p 19 Â· miR-18a-5p 9. AUTHORED

CNV_ Â· 9

reference â€” context, provenance and rarely-quoted detail

> Copy number at the arm's own locus â€” mean CN and amplification/deletion summaries.
> **9** columns Â· coverage **35%â€“35%** (median 35%) Â· **8** share the block description
> 6× percent Â· 3× real â‰¥ 0

cnv_mean_cn
ARM · REAL â‰¥ 0 · 35%
mean DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_median_cn
ARM · REAL â‰¥ 0 · 35%
median DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_pct_altered
ARM · PERCENT · 35%
percent DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_pct_amp
ARM · PERCENT · 35%
percent DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_pct_deep_del
ARM · PERCENT · 35%
Percent of samples with a deep deletion at this arm's locus. AUTHORED

cnv_pct_gain
ARM · PERCENT · 35%
percent DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_pct_loh
ARM · PERCENT · 35%
percent DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_pct_loss
ARM · PERCENT · 35%
percent DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_sd_cn
ARM · REAL â‰¥ 0 · 35%
standard deviation DERIVED
> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

GCTX_ Â· 16

reference â€” context, provenance and rarely-quoted detail

> Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware.
>
> **16** columns Â· coverage **24%â€“29%** (median 29%) Â· **15** share the block description
>
> 6× categorical Â· 4× boolean Â· 3× identifier Â· 1× constant Â· 1× count Â· 1× real â‰¥ 0

gctx_clustered
ARM BOOLEAN 29%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_context_mixed
ARM BOOLEAN 29%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_coord_source
ARM CONSTANT 29%

Provenance of the genomic-context coordinates. AUTHORED

---

gctx_host
ARM IDENTIFIER 27%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_host_region
ARM IDENTIFIER 27%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_host_type
ARM CATEGORICAL 27%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_in_exon
ARM BOOLEAN 29%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_mir_class
ARM CATEGORICAL 29%

class label DERIVED

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_mirgenedb
ARM BOOLEAN 29%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_n_loci
ARM COUNT 29%

count DERIVED

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_promoter_class
ARM CATEGORICAL 29%

class label DERIVED

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_promoter_coord_class
ARM CATEGORICAL 29%

class label DERIVED

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_promoter_gene
ARM IDENTIFIER 24%

per gene DERIVED

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_promoter_gene_type
ARM CATEGORICAL 24%

per gene DERIVED

Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)

---

gctx_promoter_host_tss_kb
ARM | REAL Â‰¥ 0 | 24%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)*

gctx_strand_rel
ARM | CATEGORICAL | 27%

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. BLOCK-LEVEL

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)*

SURROGATE_ Â· 2

reference â€” context, provenance and rarely-quoted detail

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for.

2 columns Â· coverage **0%â€“0%** (median 0%) Â· **2** share the block description
1× fraction 0â€“1 Â· 1× categorical

surrogate_corr
ARM | FRACTION 0Â€“1 | 0%

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for. BLOCK-LEVEL

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€” 3.6% of edges*

surrogate_instrument
ARM | CATEGORICAL | 0%

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for. BLOCK-LEVEL

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€” 3.6% of edges*

CNVC_ Â· 5

reference â€” context, provenance and rarely-quoted detail

CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.

5 columns Â· coverage **31%â€“31%** (median 31%) Â· **4** share the block description
2× p-value Â· 2× signed âˆ’1â€¦1 Â· 1× count

cnvc_n
ARM | COUNT | 31%

count DERIVED

> *Defined on: arms with a CNâ†’expression concordance fit (756)*

cnvc_partial_p
ARM | P-VALUE | 31%

p-value DERIVED

> *Defined on: arms with a CNâ†’expression concordance fit (756)*

cnvc_partial_q
ARM | P-VALUE | 31%

BH q for the arm's CN-coupling partial correlation. AUTHORED

cnvc_partial_rho
ARM | SIGNED Â˘'1Â€¦1 | 31%

Spearman Ï DERIVED

> *Defined on: arms with a CNâ†’expression concordance fit (756)*

cnvc_spearman_rho
ARM | SIGNED Â˘'1Â€¦1 | 31%

Spearman Ï DERIVED

> *Defined on: arms with a CNâ†’expression concordance fit (756)*

FOC_ Â· 7

reference â€” context, provenance and rarely-quoted detail

Paralog-decontaminated FOCAL-locus concordance.

7 columns Â· coverage **30%â€“30%** (median 30%) Â· **7** share the block description
2× count Â· 2× signed âˆ’1â€¦1 Â· 1× identifier Â· 1× boolean Â· 1× p-value

foc_focal_locus
ARM IDENTIFIER 30%

Paralog-decontaminated FOCAL-locus concordance. BLOCK-LEVEL

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument

---

foc_multilocus
ARM BOOLEAN 30%

Paralog-decontaminated FOCAL-locus concordance. BLOCK-LEVEL

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument

---

foc_n
ARM COUNT 30%

count DERIVED

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument

---

foc_n_loci
ARM COUNT 30%

count DERIVED

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument

---

foc_partial_p
ARM P-VALUE 30%

p-value DERIVED

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument

---

foc_partial_rho
ARM SIGNED Âˆ'1â€¦1 30%

Spearman Ï DERIVED

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument

---

foc_spearman_rho
ARM SIGNED Âˆ'1â€¦1 30%

Spearman Ï DERIVED

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument

---

MODEL_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **30%â€"30%** (median 30%)
1× count

model_n_genes
ARM COUNT AGG EDGE 30%

Which model variant produced this row. AUTHORED

Defined on: arms carrying >=1 curated HE edge in the model design (741)

COMV_ Â· 2

reference â€" context, provenance and rarely-quoted detail

Co-expression module membership and top co-expressed partner. Pure co-expression â€" the cross-state class columns are deliberately excluded from this block.

2 columns Â· coverage **15%â€"15%** (median 15%) Â· **2** share the block description
1× count Â· 1× identifier

comv_module
ARM COUNT 15%

Co-expression module membership and top co-expressed partner. Pure co-expression â€" the cross-state class columns are deliberately excluded from this block. BLOCK-LEVEL

Defined on: arms in a co-expression module (371) â€" pure co-expression; the state-class columns are excluded

---

comv_top_partner
ARM IDENTIFIER 15%

the top-ranked DERIVED

Defined on: arms in a co-expression module (371) â€" pure co-expression; the state-class columns are excluded

CLOAD_ Â· 3

reference â€" context, provenance and rarely-quoted detail

Compartment LOADING of the arm.

3 columns Â· coverage **8%â€"60%** (median 60%) Â· **3** share the block description
1× real â‰¥ 0 Â· 1× fraction 0â€"1 Â· 1× categorical

cload_dose_rpm
ARM | REAL Â‰¥ 0 | 60%

dose DERIVED

> *Defined on: arms with a compartment-loading row (1,473) â€" âš âš its source has NO PRODUCER (MH-111 ad-hoc)*

cload_lock
ARM | FRACTION 0Â€"1 | 8%

Compartment LOADING of the arm. BLOCK-LEVEL

> *Defined on: arms with a compartment-loading row (1,473) â€" âš âš its source has NO PRODUCER (MH-111 ad-hoc)*

cload_top_compartment
ARM | CATEGORICAL | 60%

the top-ranked DERIVED

> *Defined on: arms with a compartment-loading row (1,473) â€" âš âš its source has NO PRODUCER (MH-111 ad-hoc)*

HX_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· coverage **28%â€"28%** (median 28%)
1× real â‰¥ 0

hx_hly_baseline_imputed
ARM | REAL Â‰¥ 0 | 28%

â›" IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) â€" LABELLED as imputed, never presented as a measurement. AUTHORED seed_family_card.tsv

seed_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS â€” for those every 'family-level' statistic is trivially that one arm's value. Mask on `fam_single_member` first.

33 columns **1,959** rows **2** blocks median coverage **92%**

Contents â€” 2 blocks, most important first

SEED_ Â· 1

core â€” the columns findings rest on

1 columns Â· coverage **100%â€“100%** (median 100%)
1× identifier

seed_family
KEY IDENTIFIER 100%

KEY â€” the seed family itself, one row per family. AUTHORED

FAM_ Â· 32

reference â€” context, provenance and rarely-quoted detail

Seed-family structure the arm sits in â€” membership, affinity and dominance within it.

32 columns Â· coverage **4%â€“100%** (median 91%) Â· **17** share the block description
12× count Â· 8× real â‰¥ 0 Â· 7× fraction 0â€“1 Â· 2× boolean Â· 1× categorical Â· 1× identifier Â· 1× signed âˆ’1â€¦1

fam_ctx_composition
FAMILY CATEGORICAL 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%. AUTHORED

fam_ctx_context_homogeneous
FAMILY BOOLEAN 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%. AUTHORED

fam_ctx_dose_coding_host_frac
FAMILY FRACTION 0â€“1 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%. AUTHORED

fam_ctx_dose_cotranscribed_frac
FAMILY FRACTION 0â€“1 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%. AUTHORED

fam_ctx_n_distinct_coding_hosts
FAMILY COUNT 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%. AUTHORED

---

fam_cur_degree_max
FAMILY COUNT 36%

maximum DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_dominant_arm
FAMILY IDENTIFIER 92%

the dominant one Â· per arm DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_dominant_share
FAMILY FRACTION 0Â€"1 92%

The top member's share of the family's total dose. AUTHORED

---

fam_dose_hhi
FAMILY FRACTION 0Â€"1 15%

Concentration of DOSE across the family's members. âœ... **FLOOR-CORRECTED â€" verified 2026-08-19**: its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and spearman(n_members, hhi) = **+0.1896** rather than the strongly negative value a raw index would give. AUTHORED

---

fam_dose_max_log2
FAMILY REAL Â‰¥ 0 92%

dose Â· maximum DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_dose_med_log2
FAMILY REAL Â‰¥ 0 92%

dose Â· median DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_dose_total_log2
FAMILY REAL Â‰¥ 0 100%

dose Â· total DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_fame_npmid_sum
FAMILY COUNT 100%

sum DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_n_expressed
FAMILY COUNT 100%

count DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_n_guide
FAMILY COUNT 100%

count DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_n_members
FAMILY COUNT 100%

Arms in the seed family (median 1, max 22). AUTHORED

---

fam_n_members_context
FAMILY COUNT 29%

Arms in the seed family (median 1, max 22). AUTHORED

---

fam_n_model_edges
FAMILY COUNT 100%

count DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_n_passenger
FAMILY COUNT 100%

count DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

fam_n_shift_gt20_gated
FAMILY COUNT 100%

count Â· shift Â· gated DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_real_n_c10_sum
FAMILY COUNT 100%

count Â· at the âˆ’0.10 threshold Â· sum DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_real_n_scored_sum
FAMILY COUNT 100%

count Â· sum DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_redund_m_eff
FAMILY REAL â‰¥ 0 4%

Within-family redundancy â€” effective member count, over-count fraction, and the median within-family correlation. AUTHORED

---

fam_redund_median_within_rho
FAMILY SIGNED âˆ’1â€¦1 4%

Within-family redundancy â€” effective member count, over-count fraction, and the median within-family correlation. AUTHORED

---

fam_redund_overcount_frac
FAMILY FRACTION 0â€“1 4%

Within-family redundancy â€” effective member count, over-count fraction, and the median within-family correlation. AUTHORED

---

fam_seed_shift_max_gated
FAMILY FRACTION 0â€“1 13%

shift Â· maximum Â· gated DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_single_member
FAMILY BOOLEAN 100%

ââ **THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS.** For those rows every 'family-level' statistic is TRIVIALLY that one arm's value, and `fam_dominant_share` is forced to exactly 1.0. AUTHORED

---

fam_targetome_seed_med
FAMILY REAL â‰¥ 0 97%

Family-level targetome size. AUTHORED

---

fam_targetome_seq_med
FAMILY REAL â‰¥ 0 30%

Family-level targetome size. AUTHORED

---

fam_var_hhi
FAMILY FRACTION 0â€“1 14%

Seed-family structure the arm sits in â€” membership, affinity and dominance within it. BLOCK-LEVEL

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_var_max
FAMILY REAL â‰¥ 0 89%

maximum DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_var_med
FAMILY REAL â‰¥ 0 89%

median DERIVED

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*